LESSON 1.2

Stoichiometry

Answers for 5 multiple-choice and 9 structured questions

Part A - Multiple choice answers

1.2-M7	May 2024 TZ2 · P1 · Q3	A	1.2-M10	Nov 2023 TZ1 · P1 · Q3	C	1.2-M60	May 2012 TZ2 · P1 · Q2	B
1.2-M9	Nov 2023 TZ1 · P1 · Q2	D	1.2-M43	May 2015 TZ1 · P1 · Q3	B			

Part B - Structured questions markscheme

1.2-S6 May 2025 TZ3 · Paper 1B · Q3 parts (b)(i), (b)(v)

3.	(b)	(i)	<table><tr><th>*Loss in mass (g)*</th><th>*Amount of H_2O /mol*</th></tr><tr><td>5.361</td><td>0.298</td></tr></table> ✓	***Both** must be to correct sig. fig.*	1
3.	(b)	(v)	«0.60/0.10 =» 6 ✓	*Do **not** accept non-integer values.*	1

1.2-S3 May 2025 TZ2 · Paper 1B · Q3 parts (b)

3	(b)		*Limiting reagent:* HCl ***AND*** *Reason:* same initial mass of solid produces different mass of CO_2 /carbon dioxide ***OR*** different concentration/initial amount in moles of HCl produced different mass of CO_2 /carbon dioxide/gas/overall mass loss ***OR*** final mass of all mixtures would be same if $NaHCO_3$ was limiting ✓	*Accept converse argument.*	1

1.2-S17 May 2022 TZ1 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(iii), (c)(i), (c)(ii), (d)(i), (d)(iii), (e)(iv) · also 2.1, 6.1, 7.3

1.	**a**	**i**	$2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s)$ ✓	*Do not accept equilibrium arrows.* *Ignore state symbols.*	**1**

1.	**b**	**i**	$\langle\langle \frac{53.726\ g - 47.372\ g}{24.31\ g\ mol^{-1}} = \frac{6.354\ g}{24.31\ g\ mol^{-1}} \rangle\rangle$ = 0.2614 «mol» ✓		**1**

1.	**b**	**iii**	⟨⟨ $0.2614\ \text{mol x}\ (24.31\ \text{g mol}^{-1} + 16.00\ \text{g mol}^{-1}) = 0.2614\ \text{mol x}\ 40.31\ \text{g mol}^{-1}$⟩⟩ = 10.536 «g» ✓ ⟨⟨$100\ \text{x}\ \frac{\mathbf{9.569\ g}}{\mathbf{10.536g}} = 90.822$⟩⟩ = 91 «%» ✓	*Award 0.2614 mol x 40.31 g mol^{-1}.* *Accept alternative methods to arrive at the correct answer.* *Accept final answer in the range 90.5-91.5%.* ***[2]** for correct final answer.*	**2**
1.	**c**	**i**	yes ***AND*** «each Mg combines with $^2/_3$ N, so» mass increase would be $14x^2/_3$ which is less than expected increase of 16x ***OR*** 3 mol Mg would form 101g of Mg_3N_2 but would form 3 x MgO = 121 g of MgO ***OR*** 0.2614 mol forms 10.536 g of MgO, but would form 8.796 g of Mg_3N_2 ✓	*Accept Yes **AND** "the mass of N/N_2 that combines with each g/mole of Mg is lower than that of O/O_2"* *Accept YES **AND** "molar mass of nitrogen less than of oxygen".*	**1**
1.	**c**	**ii**	incomplete reaction ***OR*** Mg was partially oxidised already ***OR*** impurity present that evaporated/did not react ✓	*Accept "crucible weighed before fully cooled".* *Accept answers relating to a higher atomic mass impurity consuming less O/O_2.* *Accept "non-stoichiometric compounds formed".* *Do **not** accept "human error", "wrongly calibrated balance" or other non-chemical reasons.* *If answer to (b)(iii) is >100%, accept appropriate reasons, such as product absorbed moisture before being weighed.*	**1**
1.	**d**	**i**	«1» $Mg_3N_2(s)$ + **6** $H_2O(l)$ → **3** $Mg(OH)_2(s)$ + **2** $NH_3(aq)$ ✓		**1**
1.	**d**	**iii**	*Mg_3N_2:* -3 ***AND*** *NH_3:* -3 ✓	*Do not Accept 3 or 3-.*	**1**

1.	**e**	**iv**	nitrogen ***AND*** electron lost from first *«energy»* level/s sub-level/s-orbital ***AND*** magnesium from p sub-level/p-orbital/second *«energy»* level ***OR*** nitrogen ***AND*** electron lost from lower level «than magnesium» ✓	*Accept "nitrogen **AND** electron lost closer to the nucleus «than magnesium»".*	**1**

1.2-S21 Nov 2019 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii)

2.	**a**	**i**	4 : 1 ✓		**1**
2.	**a**	**ii**	$n_{S_2O_3^{2-}}$ = **«**0.0258 dm^3 × 0.010 mol dm^{-3} =**»** 2.58 × 10^{-4} **«**mol**»** ✓ **«** $\frac{2.58 \times 10^{-4}\ mol}{4}$ =**»** 6.45 × 10^{-5} **«**mol**»** ✓	*Award **[2]** for correct final answer.*	**2**
2.	**a**	**iii**	**«**difference in moles per dm^3 = (6.45 × 10^{-5} – 5.03 × 10^{-5}) × $\frac{1000}{300.0}$ =**»** 4.73 × 10^{-5} **«**mol dm^{-3}**»** ✓ **«**convert to mg per dm^3: 4.73 × 10^{-5} mol dm^{-3} × 32.00 g mol^{-1} × 1000 mg g^{-1} = **»** 1.51 **«**ppm/mg dm^{-3}**»** ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S26 Nov 2018 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (d) · also 7.2, 8.2

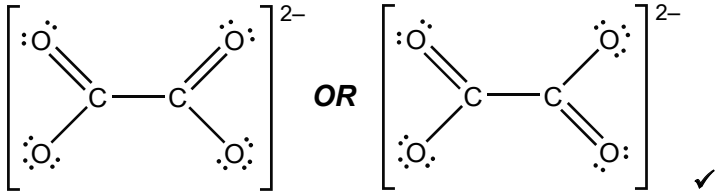
1.	**a**	**i**	n_{CuSO4} **«**$= 0.0800\,dm^3 \times 0.200\,mol\,dm^{-3}$**»** $= 0.0160\,mol$ ***AND*** n_{Fe} **«**$= \frac{3.26\,g}{55.85\,g\,mol^{-1}}$**»** $= 0.0584\,mol$ ✓ $CuSO_4$ is the limiting reactant ✓	*Do **not** award M2 if mole calculation is not shown.*	**2**
1.	**a**	**ii**	***ALTERNATIVE 1:*** **«**$0.0160\,mol \times 63.55\,g\,mol^{-1} =$**»** 1.02 **«g»** ✓ **«**$\frac{0.872\,g}{1.02\,g} \times 100 =$**»** 85.5 **«%»** ✓ ***ALTERNATIVE 2:*** **«**$\frac{0.872\,g}{63.55\,g\,mol^{-1}} =$**»** 0.0137 **«mol»** ✓ **«**$\frac{0.0137\,mol}{0.0160\,mol} \times 100 =$**»** 85.6 **«%»** ✓	*Accept answers in the range 85–86 %.* *Award **[2]** for correct final answer.*	**2**

1.	**b**	**i**	***ALTERNATIVE 1:*** $q = «80.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times 7.5\,K =» 2.5 \times 10^3$ **«J»/2.5 «kJ»** ✓ **«**per mol of $CuSO_4 = \frac{-2.5\,kJ}{0.0160\,mol} = -1.6 \times 10^2\,kJ\,mol^{-1}$**»** **«**for the reaction**»** $\Delta H = -1.6 \times 10^2$ **«**kJ**»** ✓ ***ALTERNATIVE 2:*** $q = «80.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times 7.5\,K =» 2.5 \times 10^3$ **«J»/2.5 «kJ»** ✓ **«**$n_{Cu} = \frac{0.872}{63.55} = 0.0137\,mol$**»** **«**per mol of $CuSO_4 = \frac{-2.5\,kJ}{0.0137\,mol} = -1.8 \times 10^2\,kJ\,mol^{-1}$**»** **«**for the reaction**»** $\Delta H = -1.8 \times 10^2$ **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**
1.	**b**	**ii**	density **«**of solution**»** is $1.00\,g\,cm^{-3}$ ***OR*** specific heat capacity **«**of solution**»** is $4.18\,J\,g^{-1}\,K^{-1}$/that of **«**pure**»** water ***OR*** reaction goes to completion ***OR*** iron/$CuSO_4$ does not react with other substances ✓	*The mark for "reaction goes to completion" can only be awarded if 0.0160 mol was used in part (b)(i).* *Do **not** accept "heat loss".*	**1**

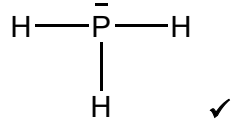
1.	**d**		*Anode (positive electrode):* $2H_2O\,(l) \rightarrow O_2\,(g) + 4H^+\,(aq) + 4e^-$ ✓ *Cathode (negative electrode):* $2H_2O\,(l) + 2e^- \rightarrow H_2\,(g) + 2OH^-\,(aq)$ ***OR*** $2H^+\,(aq) + 2e^- \rightarrow H_2\,(g)$ ✓	*Accept "$4OH^-\,(aq) \rightarrow O_2\,(g) + 2H_2O\,(l) + 4e^-$"* ***OR*** *"$Fe^{2+}\,(aq) \rightarrow Fe^{3+}\,(aq) + e^-$" for M1.* *Accept "$Fe^{2+}\,(aq) + 2e^- \rightarrow Fe\,(s)$" **OR** "$SO_4^{2-}\,(aq) + 4H^+\,(aq) + 2e^- \rightarrow 2H_2SO_3(aq) + H_2O\,(l)$" for M2.*	**2**

1.2-S35 Nov 2016 · Paper 2 · Q2 parts (c)(ii), (c)(iii), (c)(iv), (d), (e) · also 3.1, 4.1

2.	**c**	**ii**	«$n(NaOH) = \left(\frac{14.0}{1000}\right) dm^3 \times 0.100\,mol\,dm^{-3}$ =» 1.40×10^{-3} «mol» ✓		**1**
2.	**c**	**iii**	«$\frac{1}{2} \times 1.40 \times 10^{-3}$ =» 7.00×10^{-4} «mol» ✓		**1**
2.	**c**	**iv**	***ALTERNATIVE 1:*** «mass of pure hydrated ethanedioic acid in each titration = $7.00 \times 10^{-4}\,mol \times 126.08\,g\,mol^{-1}$ =» 0.0883 / 8.83×10^{-2} «g» ✓ mass of sample in each titration = «$\frac{25}{1000} \times 5.00\,g$ =» 0.125 «g» ✓ «% purity = $\frac{0.0883\,g}{0.125\,g} \times 100$ =» 70.6 «%» ✓ ***ALTERNATIVE 2:*** «mol of pure hydrated ethanedioic acid in 1 dm^3 solution = $7.00 \times 10^{-4} \times \frac{1000}{25}$ =» 2.80×10^{-2} «mol» ✓ «mass of pure hydrated ethanedioic acid in sample = $2.80 \times 10^{-2}\,mol \times 126.08\,g\,mol^{-1}$ =» 3.53 «g» ✓ «% purity = $\frac{3.53\,g}{5.00\,g} \times 100$ =» 70.6 «%» ✓ ***ALTERNATIVE 3:*** mol of hydrated ethanedioic acid (assuming sample to be pure) = $\frac{5.00\,g}{126.08\,g\,mol^{-1}} = 0.03966$ «mol» ✓ actual amount of hydrated ethanedioic acid = «$7.00 \times 10^{-4} \times \frac{1000}{25}$ =» 2.80×10^{-2} «mol» ✓ «% purity = $\frac{2.80 \times 10^{-2}}{0.03966} \times 100$ =» 70.6 «%» ✓	*Award suitable part marks for alternative methods.* *Award **[3]** for correct final answer.* *Award **[2 max]** for 50.4 % if anhydrous ethanedioic acid assumed.*	**3**

2.	**d**		 ***OR*** ✓	*Accept single negative charges on two O atoms singly bonded to C.* *Do not accept resonance structures.* *Allow any combination of dots/crosses or lines to represent electron pairs.*	**1**
2.	**e**		electrons delocalized **«**across the O–C–O system**»** ***OR*** resonance occurs ✓ 122 **«**pm**»** < C–O < 143 **«**pm**»** ✓	*Accept delocalized π-bond(s).* *No ECF from (d).* *Accept any answer in range 123 **«**pm**»** to 142 **«**pm**»**.* *Accept "bond intermediate between single and double bond" or "bond order 1.5".*	**2**

1.2-S38 May 2016 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iv), (a)(v), (b)(i), (b)(iii), (b)(iv), (c)(i), (c)(ii), (c)(iii), (d)(i) · also 3.1, 4.2, 5.1, 5.2, 6.1, 8.2

1.	**a**	**i**	 ✓	*Accept structures using dots and/or crosses to indicate bonds and/or lone pair.*	**1**
1.	**a**	**ii**	sp^3 ✓	*Do not allow **ECF** from a (i).*	**1**

1.	**a**	**iv**	non-polar ***AND*** P and H have the same electronegativity ✓	*Accept "similar electronegativities".* *Accept "polar" if there is a reference to a small difference in electronegativity and apply **ECF** in 1 a (v).*	**1**
1.	**a**	**v**	PH_3 has London «dispersion» forces ✓ NH_3 forms H-bonds ✓ H-bonds are stronger ***OR*** London forces are weaker ✓	*Accept van der Waals' forces, dispersion forces and instantaneous dipole – induced dipole forces.* *Accept "dipole-dipole forces" as molecule is polar.* *H-bonds in NH_3 (only) must be mentioned to score **[2]**.* *Do **not** award M2 or M3 if:* • *implies covalent bond is the H-bond* • *implies covalent bonds break.* *Accept "dipole-dipole forces are weaker".*	**2 max**

1.	**b**	**i**	P_4 is a molecule «comprising 4P atoms» ***AND*** 4P is four/separate «P» atoms ***OR*** P_4 represents «4P» atoms bonded together ***AND*** 4P represents «4» separate/non-bonded «P» atoms ✓		**1**

1.	**b**	**iii**	P_4: 0 ✓ $H_2PO_2^-$: +1 ✓	*Do not accept 1 or 1+ for $H_2PO_2^-$.*	**2**
1.	**b**	**iv**	oxygen gained, so could be oxidation ✓ hydrogen gained, so could be reduction ***OR*** negative charge «on product/$H_2PO_2^-$»/gain of electrons, so could be reduction ✓ oxidation number increases so must be oxidation ✓	*Award **[1 max]** for M1 and M2 if candidate displays knowledge of at least two of these definitions but does not apply them to the reaction.* *Do not award M3 for "oxidation number changes".*	**3**
1.	**c**	**i**	«$\left\langle \frac{2.478}{4 \times 30.97} \right\rangle$» $= 0.02000$ «mol» ✓		**1**
1.	**c**	**ii**	n(NaOH) = «$0.1000 \times 5.00 =$» 0.500 «mol» ***AND*** P_4/phosphorus is limiting reagent ✓	*Accept $n(H_2O) = \frac{100}{18} = 5.50$ **AND** P_4 is limiting reagent.*	**1**
1.	**c**	**iii**	amount in excess «$= 0.500 - (3 \times 0.02000)$» $= 0.440$ «mol» ✓		**1**

1.	**d**	**i**	temperature rise «$= \frac{750 \times 1.00}{0.2000 \times 1.00}$» $= 3750$ «°C/K» ✓	*Do not accept –3750.*	**1**

1.2-S43 May 2014 TZ2 · Paper 2 · Q1 parts (a), (c)(ii), (c)(iii), (g) · also 5.2

(a) $\mathrm{M(CH_3COOH)}\left(=(4\times1.01)+(2\times12.01)+(2\times16.00)\right)=60.06\,(\mathrm{g\,mol^{-1}})$;

Accept 60 (g mol^{-1}).

mass $(\mathrm{CH_3COOH})(=5.00\times1.05)=5.25\,(\mathrm{g})$;

$\frac{5.25}{60.06}=0.0874\,(\mathrm{mol})$; ***[3]***

*Award **[3]** for correct final answer.*

Accept 0.0875 (comes from using Mr = 60 g mol^{-1}).

(ii) $26.00\,(\mathrm{cm^3})$; ***[1]***

(iii) $26.00-3.00=23.00\,(\mathrm{cm^3})$;

*If other methods used, award **M1** for calculating amount of NaOH reacting with CH_3COOH.*

$0.200 \times \frac{23.00}{5.00} = 0.920\,(\text{mol dm}^{-3})$; ***[2]***

*Award **[2]** for correct final answer.*
If (ii) given as mean titre (26.5 cm^3) then ECF answer comes to 0.94 (mol dm^{-3}).

(g) ethyl ethanoate/$CH_3COOC_2H_5$/ester;
forms only weak hydrogen bonds (to water); ***[2]***
Allow "does not hydrogen bond to water" / "hydrocarbon sections too long" / OWTTE.
M2 can only be given only if M1 correct.

1.2-S44 Nov 2013 · Paper 2 · Q5 parts (a), (c), (g), (h), (i) · also 3.1, 4.1, 5.1, 9.3

5. (a) $\underline{(l)}\ N(NO_2)_3(g) + \underline{2}\ CH_3OH(l) \rightarrow \underline{2}\ N_2(g) + \underline{2}\ CO_2(g) + \underline{4}\ H_2O(l)$; ***[1]***

(c) *bonds broken:* $(6 \times 305) + (3 \times 158) = 1830 + 474 = 2304\,(\text{kJ mol}^{-1})$;
bonds made: $(2 \times 945) + (3 \times 498) = 1890 + 1494 = 3384\,(\text{kJ mol}^{-1})$;
enthalpy change: $2304 - 3384 = -1080\,(\text{kJ mol}^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*
*Award **[2 max]** for +1080 (kJ mol^{-1}).*

Accept –234 kJ mol^{-1} which arise from students assuming that 305 kJ mol^{-1} refers to the strength of a single N–O bond. Students may then take N=O from the data book value (587 kJ mol^{-1}).
bonds broken: $(3 \times 305) + (3 \times 587) + (3 \times 158) = 915 + 1761 + 474 = 3150\,(\text{kJ mol}^{-1})$
bonds made: $(2 \times 945) + (3 \times 498) = 1890 + 1494 = 3384\,(\text{kJ mol}^{-1})$
enthalpy change: $3150 - 3384 = -234\,(\text{kJ mol}^{-1})$.

*Award **[2 max]** for correct calculation of the enthalpy change of reaction for the equation in part (a), which gives –2160 (kJ mol^{-1}).*
*Award **[1]** if the final answer is not –2160 but the candidate has correctly calculated the bonds broken in trinitramide as 2304 (kJ mol^{-1}).*

(g) (N–N bond in) trinitramide is longer/nitrogen (gas) is shorter / 0.145 nm in trinitramide versus 0.110 nm in nitrogen;

(h) $106^\circ - 108^\circ$;
Accept $< 109^\circ$.

Any two for ***[2 max]***.
4 (negative) charge centres/electron pairs/electron domains around central nitrogen;
central nitrogen has a lone/non-bonding pair;
lone/non-bonding pairs repel more than bonding pairs;
molecule will be (trigonal/triangular) pyramidal;
(negative) charge centres/electron pairs/electron domains will be tetrahedrally arranged/orientated/ have tetrahedral geometry; ***[3 max]***
Do not apply ECF.

(i) polar;
net dipole moment present in molecule / unsymmetrical distribution of charge / polar bonds do not cancel out / centre of negatively charged oxygen atoms does not coincide with positively charged nitrogen atom; ***[2]***
Marks may also be awarded for a suitably presented diagram showing net dipole moment.
Do not accept "unsymmetrical molecule".
Apply ECF from part (h).

LESSON 2.1

Ionization Energy

Answers for 11 multiple-choice and 4 structured questions

Part A - Multiple choice answers

2.1-M5	May 2025 TZ2 · P1A · Q4	**A**	**2.1-M14**	May 2024 TZ2 · P1 · Q9	**B**	**2.1-M23**	May 2021 TZ1 · P1 · Q6	**A**
2.1-M19	Nov 2022 · P1 · Q4	**C**	**2.1-M22**	Nov 2021 · P1 · Q5	**D**	**2.1-M38**	May 2017 TZ1 · P1 · Q5	**B**
2.1-M10	Nov 2024 · P1 · Q7	**D**	**2.1-M32**	Nov 2018 · P1 · Q7	**B**	**2.1-M51**	May 2014 TZ1 · P1 · Q5	**C**
2.1-M13	May 2024 TZ2 · P1 · Q6	**A**	**2.1-M57**	May 2012 TZ1 · P1 · Q8	**D**			

Part B - Structured questions markscheme

2.1-S3 May 2025 TZ2 · Paper 2 · Q2 parts (b)(iii), (b)(vii), (b)(viii), (c)(i), (c)(ii), (c)(iii) · also 6.3, 7.2, 9.4

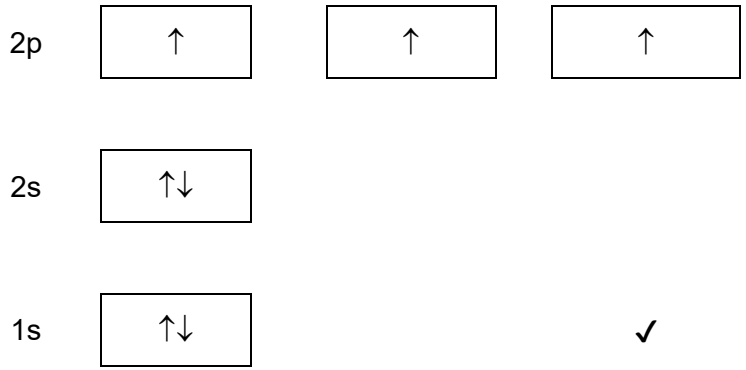
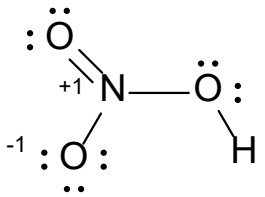
2	**(b)**	**(iii)**	lattice enthalpy ✓		**1**

2	**(b)**	**(vii)**	***ALTERNATIVE 1:*** place thallium in a solution of copper sulfate ✓ if reaction occurs then thallium is more reactive ***OR*** if no reaction occurs then copper is more reactive ✓ ***ALTERNATIVE 2:*** place copper in a solution of thallium sulfate ✓ if reaction occurs then copper is more reactive ***OR*** if no reaction occurs then thallium is more reactive ✓		**2**

2	**(b)**	**(vii i)**	*Anode:* oxygen ***AND*** water is more easily oxidized than sulfate ion ✓ *Cathode:* thallium ***AND*** thallium is more easily reduced than water ✓		**2**
2	**(c)**	**(i)**	limit of convergence «of continuum» ✓	*Accept high energy limit instead of limit of convergence.*	**1**

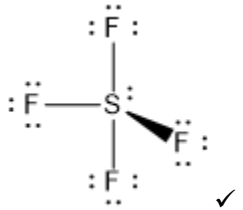
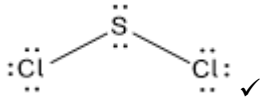
2	**(c)**	**(ii)**	IE ⟨⟨= $\frac{589\ x\ 10^3\ Jmol^{-1}}{6.02\ x\ 10^{23}\ mol^{-1}}$⟩⟩ = 9.78 x 10^{-19}«J $atom^{-1}$» ✓ frequency ⟨⟨= $\frac{E}{h} = \frac{9.78\ x\ 10^{-19}\ J}{6.63\ x\ 10^{-34}\ Js}$⟩⟩ = 1.48 x 10^{15} «s^{-1}» ✓ wavelength ⟨⟨= $\frac{c}{f} = \frac{3.00x10^8 m\ s^{-1}}{1.48\ x\ 10^{15}\ s^{-1}}$⟩⟩ = 2.03 x 10^{-7} «m» / 203 «nm» ✓	*Award **[3]** for correct final answer.*	**3**
2	**(c)**	**(iii)**	nuclear charge of Tl less than that of Pb / increasing «effective» nuclear charge from Tl to Pb ✓ shielding effect/occupied energy levels stay the same ✓	*Do **not** accept the atomic radius of lead is less than that of thallium for M1.* *Do **not** accept that the loss of an electron from thallium leaves a complete subshell.*	**2**

2.1-S7 May 2022 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii) · also 3.1, 4.1

6.	**a**	**i**	 ✓	*Accept **all** 2p electrons pointing downwards.* *Accept half arrows instead of full arrows.*	**1**
6.	**a**	**ii**	 bonds and non-bonding pairs correct ✓ formal charges correct ✓	*Accept dots, crosses or lines to represent electron pairs.* *Do **not** accept resonance structures with delocalised bonds/electrons.* *Accept + and – sign respectively.* *Do not accept a bond between nitrogen and hydrogen.* *For an incorrect Lewis structure, allow ECF for non-zero formal charges*.	**2**
6.	**a**	**iii**	*Any three of:* two N-O same length/order ✓ delocalization/resonance ✓ N-OH longer «than N-O» ***OR*** N-OH bond order 1 ***AND*** N-O bond order 1½ ✓	*Award **[2 max]** if bond strength, rather than bond length discussed.* *Accept N-O between single and double bond **AND** N-OH single bond.*	**3**

2.1-S13 May 2021 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i), (c), (d) · also 3.1, 5.2

2.	**a**	**ii**	P has «three» unpaired electrons in 3p sub-level ***AND*** S has one full 3p orbital «and two 3p orbitals with unpaired electrons» ***OR*** P: $[Ne]3s^2 3p_x^1 3p_y^1 3p_z^1$ ***AND*** S: $[Ne]3s^2 3p_x^2 3p_y^1 3p_z^1$ ✓ repulsion between paired electrons in sulfur **«**and therefore easier to remove**»** ✓	*Accept orbital diagrams for 3p sub-level for M1. Ignore other orbitals or sub-levels.* *Accept "removing electron from S gives more stable half-filled sub-level" for M2.*	**2**
2.	**b**	**i**	*Cr:* [Ar] $4s^1 3d^5$ ✓ *Cr^{3+}:* [Ar] $3d^3$ ✓	*Accept "[Ar] $3d^5 4s^1$".* *Accept "[Ar] $3d^3 4s^0$".* *Award **[1 max]** for two correct full electron configurations "$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ **AND** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$".* *Award **[1 max]** for $4s^1 3d^5$ **AND** $3d^3$.*	**2**

<table>
<tr>
<td>2.</td>
<td>c</td>
<td></td>
<td>
<table>
<tr><th>Species</th><th>SF_4</th><th>SCl_2</th></tr>
<tr><td>Lewis structure</td><td>  ✓</td><td>  ✓</td></tr>
<tr><td>Molecular geometry</td><td>saw horse/
see saw ✓</td><td>bent/V
shaped/angular ✓</td></tr>
</table>
</td>
<td>SF_4/SCl_2 structure does not have to be 3-D for mark.

Penalize missing lone pairs of electrons on halogens once only.

Accept any combination of dots, lines or crosses for bonds/lone pairs.

Accept “non-linear” for SCl_2 molecular geometry.

Award [1] for two correct electron domain geometries, e.g. trigonal bipyramidal for SF_4 and tetrahedral for SCl_2.</td>
<td>4</td>
</tr>
<tr>
<td>2.</td>
<td>d</td>
<td></td>
<td>H_2O forms hydrogen bonding «while SCl_2 does not» ✓

SCl_2 «much» stronger London/dispersion/«instantaneous» induced dipole-induced dipole forces ✓

Alternative 1:
H_2O less volatile AND hydrogen bonding stronger «than dipole–dipole and dispersion forces» ✓

Alternative 2:
SCl_2 less volatile AND effect of dispersion forces «could be» greater than hydrogen bonding ✓</td>
<td>Ignore reference to Van der Waals.

Accept “SCl_2 has «much» larger molar mass/electron density” for M2.</td>
<td>3</td>
</tr>
</table>

2.1-S31 May 2012 TZ1 · Paper 2 · Q8 parts (b)(i), (b)(ii)

(b) (i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$;
4s; ***[2]***

(ii) 4th electron removed from 3d **and** 5th electron removed from 3p;
10th electron removed from 3p **and** 11th electron removed from 3s;

Accept either of the following answers for the third mark:
electrons removed from lower energy level / energy level closer to nucleus are attracted more strongly;
greater effective nuclear charge / s electrons more penetrating; ***[3 max]***

LESSON 3.1

VSEPR Theory

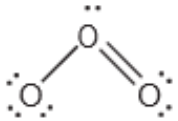
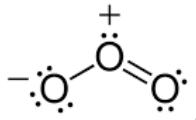
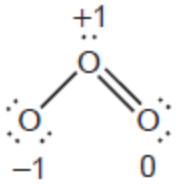
Answers for 7 multiple-choice and 3 structured questions

Part A - Multiple choice answers

3.1-M2	May 2025 TZ1 · P1A · Q10	**A**	**3.1-M14**	May 2023 TZ2 · P1 · Q12	**C**	**3.1-M40**	Nov 2012 · P1 · Q12	**B**
3.1-M6	May 2024 TZ1 · P1 · Q11	**C**	**3.1-M24**	Nov 2018 · P1 · Q9	**C**			
3.1-M16	Nov 2022 · P1 · Q11	**D**	**3.1-M27**	May 2018 TZ2 · P1 · Q11	**B**			

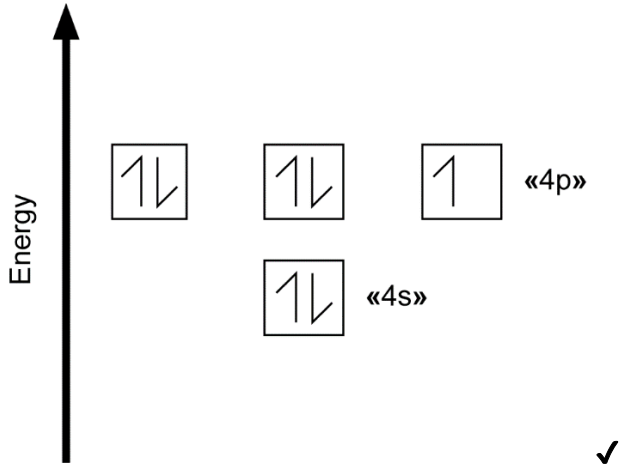
Part B - Structured questions markscheme

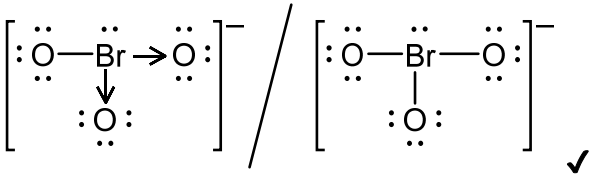
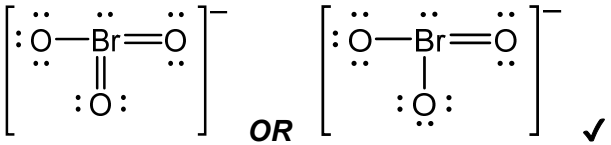
3.1-S5 May 2024 TZ2 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d) · also 2.1, 4.1, 4.2

2.	**(a)**		*Electron configuration:* $1s^2 2s^2 2p^4$ ✓ *Unpaired electrons:* 2 ✓		**2**
2.	**(b)**	**(i)**	 ✓	*Accept any combination of dots, crosses and lines for electron pairs, including an arrow for the coordinate bond.* *Do **not** accept delocalized structures.* *Accept*  .	**1**
2.	**(b)**	**(ii)**	 formal charge of central O = +1 ✓ formal charge of single bonded O = –1 ***AND*** double bonded O = 0 ✓		**2**

2.	**(b)**	**(iii)**	bent/angular/V-shaped ✓ any estimated value in the range *110 ° – 119 °* ✓	*Accept a bent diagram for M1.*	**2**
2.	**(b)**	**(iv)**	sp^2 ✓		**1**
2.	**(c)**		**«**both equal and**»** any estimated value in the range 122–147 **«**pm**»** ✓ resonance ***OR*** delocalization ***OR*** bond order is 1.5 ✓		**2**
2.	**(d)**		shorter wavelength radiation/UV needed to break the bond in O_2 ✓ O_2 has stronger bond ✓		**2**

3.1-S14 Nov 2018 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 2.1, 6.2

3.	**a**	**i**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$ ***OR*** $[Ar]\ 4s^2\ 3d^{10}\ 4p^5$ ✓	*Accept 3d before 4s.*	**1**
3.	**a**	**ii**	 ✓	*Accept double-headed arrows.*	**1**

3.	**b**	**i**	*Structure I – follows octet rule:*  ✓ *Structure II – does not follow octet rule:*  ***OR*** ✓	*Accept dots, crosses or lines to represent electron pairs.*	**2**
3.	**b**	**ii**	**«**structure I**»** formal charge on Br = +2 ***OR*** **«**structure II**»** formal charge on Br = 0/+1 ✓ structure II is preferred ***AND*** it produces formal charge closer to 0 ✓	*Ignore any reference to formal charge on oxygen.*	**2**

3.	**c**		*Geometry:* trigonal/pyramidal ✓ *Reason:* three bonds ***AND*** one lone pair ***OR*** four electron domains ✓ *O–Br–O angle:* 107° ✓	*Accept "charge centres" for "electron domains".* *Accept answers in the range 104–109°.*	**3**
3.	**d**	**i**	$BrO_3^-(aq) + 6e^- + 6H^+(aq) \rightarrow Br^-(aq) + 3H_2O(l)$ correct reactants and products ✓ balanced equation ✓	*Accept reversible arrows.*	**2**
3.	**d**	**ii**	$BrO_3^-(aq) + 6Fe^{2+}(aq) + 6H^+(aq) \rightarrow Br^-(aq) + 3H_2O(l) + 6Fe^{3+}(aq)$ ✓		**1**

3.1-S24 May 2013 TZ2 · Paper 2 · Q5 parts (a)(iii), (a)(iv), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 5.1, 9.2

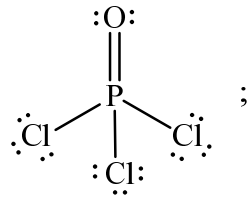
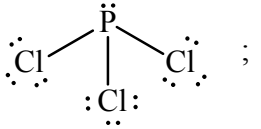
(iii) heat/enthalpy change/required/absorbed when 1 mol of a compound is formed from its elements in their standard states/at 100 kPa/10^5 Pa/1 bar; ***[1]***

Allow 1.01×10^5 *Pa / 101 kPa / 1 atm.*

Allow under standard conditions or standard temperature and pressure. Temperatures not required in definition, allow if quoted (for example, 298 K/25 °C – most common) but pressure value must be correct if stated.

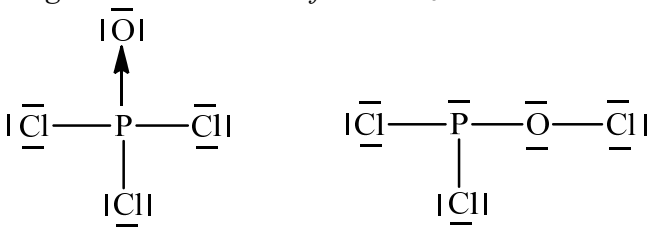
(iv) $\left(\Delta H^{\ominus} = [(2)(-288.1)] - [(2)(-542.2)]\right) = (+)508.2\ (\text{kJ mol}^{-1})$; ***[1]***

(b) (i)

	$POCl_3$	PCl_3
Lewis (electron dot) Structure	 ; *Accept legitimate alternatives for $POCl_3$ as shown below*.*	 ;
	Allow any combination of dots/crosses or lines to represent electron pairs. *Penalise missing lone pair on P in PCl_3 but penalize missing lone pairs once only on terminal O or Cl atoms.*	
Shape	tetrahedral;	trigonal/triangular pyramidal; *Do not allow tetrahedral.* *Do not allow just pyramidal.*
	Do not allow ECF from Lewis structures with incorrect number of negative charge centres. *Only allow shapes based on legitimate structures below.*	

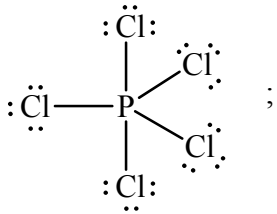
[4]

**Legitimate alternatives for $POCl_3$:*



(ii) allow any bond angle in the range 100° to less than 109° (experimental value is 100°);
due to four negative charge centres/four electron pairs/four electron domains (one of which is a lone pair)/tetrahedral arrangement of electron pairs/domains;
extra repulsion due to lone pair electrons / lone pairs occupy more space (than bonding pairs) so Cl–P–Cl bond angle decreases from 109.5° / *OWTTE*;

[3]

(c) (i)  ; ***[1]***

Allow any combination of dots/crosses or lines to represent electron pairs.
Do not penalise missing lone pairs on Cl if already penalised in (b)(i).

(ii) trigonal/triangular bipyramidal; ***[1]***

Do not allow ECF from Lewis structures with incorrect number of negative charge centres.

(iii) 120° **and** 90°/180°; ***[1]***

Ignore other bond angles such as 240° and 360°.
Apply list principle if some correct and incorrect angles given.

(iv)

	Isomer 1	Isomer 2	Isomer 3
Structure	both Br's in axial positions, all Cl's in equatorial positions	both Br's in equatorial positions, two Cl's in axial positions, one Cl in equatorial position	one Br in equatorial position, one Br in axial position, one Cl in axial position and two Cl's in equatorial positions
Molecular polarity	non-polar	polar	polar

[3]

*Award **[1]** for correct structure **and** molecular polarity.*
*Award **[1 max]** for correct representations of all three isomers.*
Lone pairs not required.

LESSON 4.1

Sigma & Pi Bonds and Delocalization

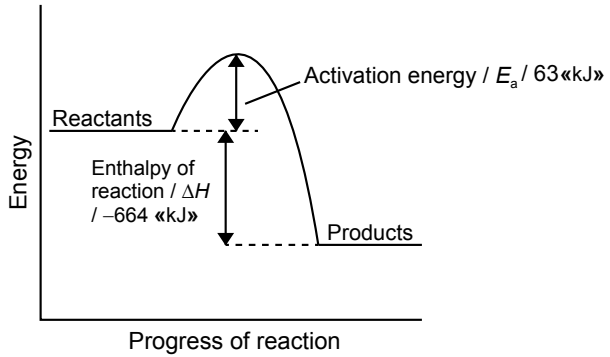
Answers for 7 multiple-choice and 2 structured questions

Part A - Multiple choice answers

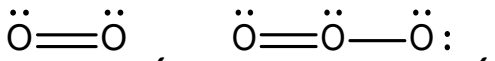
4.1-M2	Nov 2024 · P1 · Q13	**C**	**4.1-M15**	Nov 2020 · P1 · Q9	**D**	**4.1-M31**	May 2017 TZ2 · P1 · Q15	**D**
4.1-M9	Nov 2022 · P1 · Q9	**B**	**4.1-M25**	May 2018 TZ2 · P1 · Q13	**B**			
4.1-M13	May 2021 TZ1 · P1 · Q12	**D**	**4.1-M26**	Nov 2017 · P1 · Q14	**C**			

Part B - Structured questions markscheme

4.1-S11 May 2016 · Paper 2 · Q3 parts (b)(i), (c)(i), (c)(ii) · also 8.1

3.	**b**	**i**	 products lower than reactants ***AND*** enthalpy of reaction correctly marked and labelled with name or value ✓ activation energy correctly marked and labelled with name or value ✓	*Accept other clear ways of indicating energy/ enthalpy changes.*	**2**
3.	**c**	**i**	**ALTERNATIVE 1:** σ-bond from N to N ***AND*** from N to O ✓ π-bond from N to N ✓ <u>delocalized</u> π-bond/π-electrons «extending over the oxygen and both nitrogens» ✓ **ALTERNATIVE 2:** both have 2 σ-bonds «from N to N and from N to O» ***AND*** π-bond from N to N ✓ one structure has second π-bond from N to N and the other has π-bond from N to O ✓ delocalized π-bond/π-electrons ✓	*Award **[1 max]** if candidate has identified both/either structure having 2 σ-bonds and 2 π-bonds.*	**3**
3.	**c**	**ii**	more than one possible position for a multiple/π-/pi- bond ✓	*Accept "more than one possible Lewis structure".* *Accept reference to delocalisation if M3 not awarded in c (i).* *Accept reference to fractional bond orders.*	**1**

4.1-S4 Nov 2019 · Paper 2 · Q1 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 3.1

1.	**a**		 ✓	*Coordinate bond may be represented by an arrow.* *Do **not** accept delocalized structure for ozone.*	**2**
1.	**b**		resonance **«**structures**»** ***OR*** delocalization of **«**the double/pi bond**»** electrons ✓ 121 **«**pm**»** < length < 148 **«**pm**»** ✓	*Accept any length between these two values.*	**2**
1.	**c**		any value from 110°–119° ✓		**1**
1.	**d**		**«**bond**»** in O_2 stronger than in O_3 ✓ ozone absorbs lower frequency/energy **«**radiation than oxygen**»** ***OR*** ozone absorbs longer wavelength **«**radiation than oxygen**»** ✓	*Accept ozone **«**layer**»** absorbs a range of frequencies.*	**2**
1.	**e**	**i**	steps 1 ***AND*** 3 ✓		**1**

1.	**e**	**ii**	***ALTERNATIVE 1:*** for oxygen: $E = \text{«}\frac{498\,000\ \text{J mol}^{-1}}{6.02\times10^{23}\ \text{mol}^{-1}}\text{»}\ 8.27\times10^{-19}$ **«J»** ✓ $\lambda = \text{«}\frac{6.63\times10^{-34}\ \text{J s}\times3.00\times10^{8}\ \text{m s}^{-1}}{8.27\times10^{-19}\ \text{J}}\text{»}\ 2.40\times10^{-7}$ **«m»** ✓ ***ALTERNATIVE 2:*** for ozone: similar calculation using 200 < bond enthalpy < 400 for ozone, such as $E = \text{«}\frac{300\,000\ \text{J mol}^{-1}}{6.02\times10^{23}\ \text{mol}^{-1}}\text{»}\ 4.98\times10^{-19}$ **«J»** ✓ $\lambda = \text{«}\frac{6.63\times10^{-34}\ \text{J s}\times3.00\times10^{8}\ \text{m s}^{-1}}{4.98\times10^{-19}\ \text{J}}\text{»}\ 3.99\times10^{-7}$ **«m»** ✓	*Award **[2]** for correct final answer.*	**2**

LESSON 4.2

Hybridization

Answers for 4 multiple-choice and 1 structured questions

Part A - Multiple choice answers

4.2-M2	May 2025 TZ3 · P1A · Q10	**B**	**4.2-M10**	May 2019 TZ1 · P1 · Q13	**A**
4.2-M3	Nov 2024 · P1 · Q12	**A**	**4.2-M14**	Nov 2017 · P1 · Q13	**A**

Part B - Structured questions markscheme

4.2-S2 Nov 2022 · Paper 2 · Q2 parts (b), (d), (e)(ii) · also 2.1, 4.1

2.	**(b)**		nine/9 ✓		**1**
2.	**(d)**		«bond in ring is» shorter ***AND*** more electrons are shared ***OR*** «bond in ring is» shorter ***AND*** partial double/multiple bonding/bond order 1.5 ✓		**1**
2.	**(e)**	**(ii)**	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ / $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ ***OR*** $[Ar]4s^1 3d^{10}$ / $[Ar]3d^{10} 4s^1$ ✓		**1**

LESSON 4.3

Covalent Network Structures

Answers for 1 multiple-choice and 2 structured questions

Part A - Multiple choice answers

4.3-M10 May 2014 TZ2 · P1 · Q13 **A**

Part B - Structured questions markscheme

4.3-S6 May 2018 TZ2 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b), (c)

1.	**a**	**i**	consists of single/one sheet/layer **«**of carbon atoms**»** ✓ graphene has no density measurement ***OR*** graphene has no distance between layers data ***OR*** graphene has large specific surface area **«**compared to graphite**»** ✓	*Do **not** accept "sp^2" alone without reference to single/one sheet/layer.* *Accept "thickness of one atom" **OR** "consists of a plane" for M1.*	**2**
1.	**a**	**ii**	*Any one of these alternatives:* ***ALTERNATIVE 1*** **«**$\frac{1.3\times10^{11}}{76\times10^{6}}$**»** $1.7 \times 10^3/1711$ ✓ ***ALTERNATIVE 2*** $1600 \times 76 \times 10^6 = 1.2 \times 10^{11}$ **«**is less than tensile strength of graphene**»** ✓ ***ALTERNATIVE 3*** $\frac{1.3\times10^{11}}{1600} = 8.1 \times 10^7$ **«**is greater than upper end of tensile strength for graphite**»** ✓	*Accept any value in the range 1700–27 083. Answer may be expressed in scientific notation or otherwise.* *Accept any value calculated which is less than the graphene tensile strength based on a value chosen from within the 4.8–76 $\times$ 10^6 range.*	**1**

1.	**b**		smaller/zero ✓ no delocalized electrons/electrons are bound/electrons not free to move/electrons not free to roam ***OR*** localized electrons **«**in sigma bonds**»** ***OR*** large band gap ✓	*Accept "diamond is a dielectric"* ***OR*** *"diamond does* ***not*** *conduct electricity" for M2.* *Award* ***[1 max]*** *for just "immobile/less mobile".* *Award* ***[2]*** *for "electrons immobile* ***«****in diamond****»*** *due to the large band gap"* ***OR*** *"electrons* ***«****in diamond****»*** *immobile since electrons are localized* ***«****in the sigma bonds****»****".*	**2**
1.	**c**		shorter bonds in graphene ***OR*** bonds in graphene intermediate between single and double ***OR*** bond order in graphene is 1.33 ***OR*** delocalization creates stronger bonds ***OR*** shorter bonds are stronger ✓ stronger/shorter bonds require higher temperature/faster thermal motion to be altered ***OR*** stronger/shorter bonds require greater energy to be broken ✓		**2**

4.3-S10 May 2011 TZ1 · Paper 2 · Q6 parts (a), (b)(i), (b)(ii), (c), (d), (e) · also 4.1, 4.2

6. (a) *Award **[2 max]** for three of the following features:*

Bonding

*Graphite **and** C_{60} fullerene:* covalent bonds **and** van der Waals'/London/dispersion forces;
Diamond: covalent bonds (and van der Waals'/London/dispersion forces);

Delocalized electrons

*Graphite **and** C_{60} fullerene:* delocalized electrons;
Diamond: no delocalized electrons;

Structure

Diamond: network/giant structure / macromolecular / three-dimensional structure **and** *Graphite:* layered structure / two-dimensional structure / planar;
C_{60} fullerene: consists of molecules / spheres made of atoms arranged in hexagons/pentagons;

Bond angles

Graphite: 120° **and** *Diamond:* 109°;
C_{60} fullerene: bond angles between $109–120^{o}$;
*Allow Graphite: sp^2 **and** Diamond: sp^3.*
*Allow C_{60} fullerene: sp^2 **and** sp^3.*

Number of atoms each carbon is bonded to

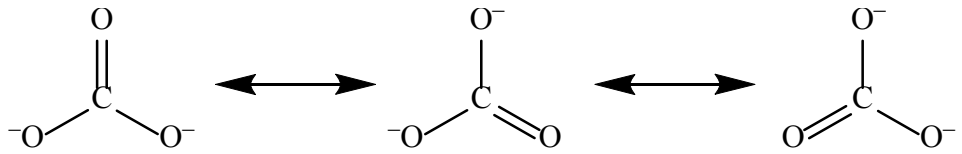
*Graphite **and** C_{60} fullerene:* each C atom attached to 3 others;
Diamond: each C atom attached to 4 atoms / tetrahedral arrangement of C (atoms); ***[6 max]***

(b) (i) network/giant structure / macromolecular;
each Si bonded covalently to 4 oxygen atoms **and** each O atom bonded covalently to 2 Si atoms / single covalent bonds; ***[2]***
*Award **[1 max]** for answers such as network-covalent, giant-covalent or macromolecular-covalent.*
Both M1 and M2 can be scored by a suitable diagram.

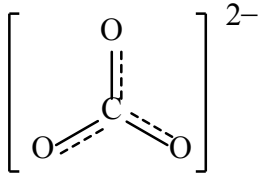
(ii) *Silicon dioxide:* strong/covalent bonds in network/giant structure/macromolecule;
Carbon dioxide: weak/van der Waals'/dispersion/London forces between molecules; ***[2]***

(c) triple (covalent) bond;
one electron pair donated by oxygen to carbon atom / dative (covalent)/coordinate (covalent) bond; ***[2]***
*Award **[1 max]** for representation of* C≡O .
Award [2] if CO shown with dative covalent bond

(d) delocalization/spread of pi/π electrons over more than two nuclei;
equal bond order/strength/length / spreading charge (equally) over all three oxygens;
gives carbonate ion a greater stability/lower potential energy; ***[3]***
M3 can be scored independently.
Accept suitable labelled diagrams for M1 and M2 e.g.



OR



Do not penalize missing brackets on resonance structure but 2– charge must be shown.
Allow –2 for charge on resonance structure.

(e) mixing/combining/merging of (atomic) orbitals to form new orbitals (for bonding);
Allow molecular or hybrid instead of new.
Do not allow answers such as changing shape/symmetries of atomic orbitals.

Carbon dioxide: sp;
Diamond: sp^3;
Graphite: sp^2;
Carbonate ion: sp^2; ***[5]***

LESSON 5.1

Molecular Polarity

Answers for 2 multiple-choice and 0 structured questions

Part A - Multiple choice answers

5.1-M2	May 2024 TZ2 · P1 · Q15	A	5.1-M3	Nov 2020 · P1 · Q10	A

LESSON 5.2

Intermolecular Forces

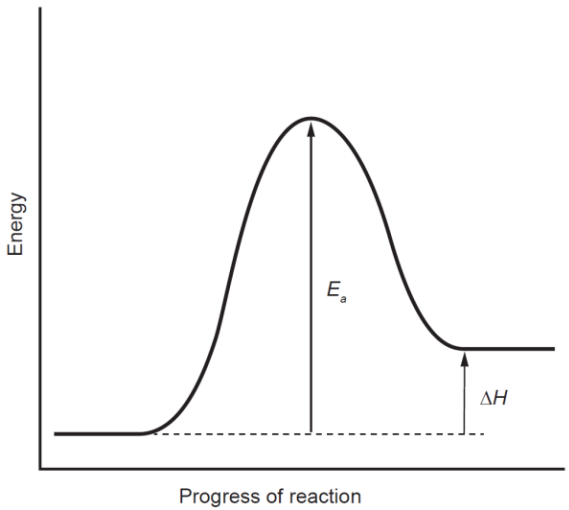
Answers for 5 multiple-choice and 2 structured questions

Part A - Multiple choice answers

5.2-M3	Nov 2023 TZ1 · P1 · Q11	A	5.2-M12	Nov 2018 · P1 · Q11	D	5.2-M23	Nov 2013 · P1 · Q11	A
5.2-M4	May 2023 TZ2 · P1 · Q10	B	5.2-M20	May 2015 TZ2 · P1 · Q11	A			

Part B - Structured questions markscheme

5.2-S3 May 2024 TZ2 · Paper 2 · Q5 parts (b), (c)(i), (c)(ii) · also 8.1, 9.2

5.	**(b)**		both have **«**similar**»** London/dispersion forces **«**due to having same number of electrons/similar M_r**»** ✓ ethenone has dipole-dipole forces **«**and carbon dioxide does not**»** ✓		**2**
5.	**(c)**	**(i)**	**«**$\Delta H° = \sum \Delta H°_{f\ products} - \sum \Delta H°_{f\ reactants}$ **»** = -74.0 - 87.2 + 248 ✓ = **«**+**»**86.8 **«**kJ**»** ✓	*Award **[2]** for correct final answer.* *Award **[1]** for –86.8 **«**kJ**»**.*	**2**
5.	**(c)**	**(ii)**	 endothermic graph ***AND*** labelled arrow/line showing ΔH ✓ labelled arrow/line showing E_a ✓		**2**

5.2-S1 May 2025 TZ3 · Paper 2 · Q7 parts (b)(i), (b)(ii), (c) · also 4.3

7.	**(b)**	**(i)**	O of C=O with H of H-N on opposite chain ✓	*bonds.*	**1**
7.	**(b)**	**(ii)**	H that is «covalently» bonded to a «highly» electronegative atom ✓ is attracted to the electronegative atom of neighbouring molecule ✓	*Accept a labelled diagram for both marks.* *Accept "F, O or N" for "electronegative atom".*	**2**
7.	**(c)**		covalent bonds between silicon atoms ✓ London/dispersion forces between poly(ethene) «chains/molecules» ✓ covalent bonds «much» stronger than London/dispersion/intermolecular forces «hence Si has higher melting point» ✓	*M3: Do **not** accept "bonding in silicon is stronger than poly(ethene)" without named bonding/forces.*	**3**

LESSON 5.3

Separation Techniques

Answers for 1 multiple-choice and 1 structured questions

Part A - Multiple choice answers

5.3-M1 May 2025 TZ1 · P1A · Q12 **C**

Part B - Structured questions markscheme

5.3-S1 May 2025 TZ1 · Paper 1B · Q2 parts (a), (e)

2.	**(a)**		*Any four of:* use magnet to remove iron/Fe ✓ add «excess» water to dissolve salt/NaCl ✓ filter/wash sand «into saltwater filtrate»✓ allow/heat sand to dry ✓ boil off/evaporate water from salt ***OR*** allow salt to dry overnight ✓	*Accept other reasonable orders to separations and names or formulas.* *Accept references to NaCl(aq) for M2.* *Do **not** accept responses that include sifting, separating with a sieve, **OR** by particle size.* *Accept "distill off water from salt" for M5.*	**4 max**
2.	**(e)**		*Separation method:* «fractional» distillation ✓ *Physical property:* boiling point ✓		**2**

LESSON 6.1

Redox Reactions & Oxidation Numbers

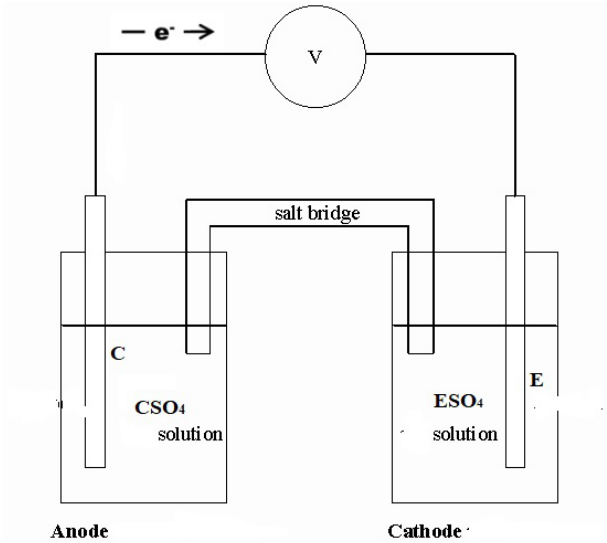
Answers for 9 multiple-choice and 2 structured questions

Part A - Multiple choice answers

6.1-M6	Nov 2024 · P1 · Q28	**B**	**6.1-M13**	May 2022 TZ1 · P1 · Q28	**D**	**6.1-M25**	May 2018 TZ1 · P1 · Q28	**C**
6.1-M10	May 2024 TZ2 · P1 · Q32	**A**	**6.1-M23**	Nov 2018 · P1 · Q8	**D**	**6.1-M32**	May 2017 TZ1 · P1 · Q29	**C**
6.1-M12	May 2023 TZ2 · P1 · Q28	**C**	**6.1-M24**	Nov 2018 · P1 · Q28	**D**	**6.1-M34**	May 2017 TZ2 · P1 · Q29	**D**

Part B - Structured questions markscheme

6.1-S3 May 2023 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b), (c) · also 6.3, 7.1, 7.2

2	**(a)**	**(i)**	+2/II ✓	*Do **not** accept A^{2+}, A^{+2}, 2 **OR** 2+.*	**1**
2.	**(a)**	**(ii)**	C A B D E ✓	*Do **not** accept reverse order unless the answer makes it clear which is the most reactive.*	**1**
2.	**(b)**		 two half cells with electrodes, solutions, salt bridge and connecting wire ✓ C as anode ***AND*** E as cathode ✓ CSO_4 ***AND*** ESO_4 «solutions» labelled ✓ electron flow shown «on wire» from more reactive to less reactive metal ✓	*Voltmeter not needed.* *Accept salt bridge even if not labelled.* *Do **not** award M1 if a battery is included.* *Do **not** award M1 if liquid level not shown in container, or if electrodes and ends of salt bridge are not in solutions.* *Accept salt of C/C^{2+}«(aq)» **AND** salt of E/E^{2+}«(aq)» for M3.*	**4**
2.	**(c)**		copper/Cu ***OR*** silver/Ag ✓	*Accept lead/Pb, tin/Sn or nickel/Ni.*	**1**

6.1-S14 Nov 2012 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii) · also 6.2, 7.2

4. (a) (i) *Oxidation:* loss of electrons **and** *Reduction:* gain of electrons; ***[1]***

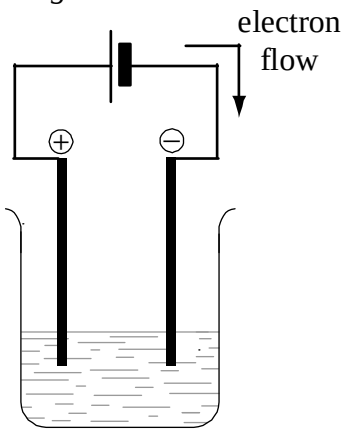
(ii) As_2O_3: +3;
NO_3^-: +5;
H_3AsO_3: +3;
N_2O_3: +3; ***[4]***
Penalize incorrect notation e.g. III, V, 3+, 5+, 3, 5 once only.

(iii) *Oxidizing agent:* substance reduced / removes electrons from another substance / causes some other substance to be oxidized / *OWTTE* **and** *Reducing agent*: substance oxidized / gives electrons to another substance / causes some other substance to be reduced / *OWTTE*; ***[1]***
*Accept Oxidizing agent: electron/e/e^- acceptor / causes oxidation / oxidation number/state decreases **and** Reducing agent: electron/e/e^- donor / causes reduction / oxidation number/state increases.*

(iv) $As_2O_3(s) + 2NO_3^-(aq) + 2H^+(aq) + 2H_2O(l) \rightarrow 2H_3AsO_4(aq) + N_2O_3(aq)$
correct coefficients for As_2O_3, H_3AsO_4 **and** NO_3^-, N_2O_3;
correct balanced equation;
Ignore state symbols.
M1 must be correct to award M2.

Oxidizing agent: $NO_3^-(aq)$/nitrate **and** *Reducing agent:* $As_2O_3(s)$/arsenic(III) oxide; ***[3]***
Accept HNO_3(aq)/nitric acid.
Accept arsenic oxide.
Species must be fully correct to score M3.
Ignore state symbols.

(b) (i) *Diagram to show*:



Labels are not required.
one container, two electrodes, battery (and electrolyte);

Allow + and – for representation of battery (could be long and short lines for example) but M1 is not scored if a voltmeter/V if shown or labelled.
Ignore designation of electrodes (e.g. do not penalize Cu electrodes).

correct direction of electron flow (from negative pole to positive pole); ***[2]***
Allow arrow without stating e^- explicitly.
To score M2, the polarity of the battery or the cathode and anode must be shown.
If a voltmeter/V is shown, M1 is not awarded but M2 may be scored if the cathode and anode are identified with the correct direction of electron flow.

correct state symbols in **all** equations;
M3 can only be scored if the correct species are given in M1 and M2 (i.e. do not award ECF from M1 and M2 for incorrect species).

Observation at positive electrode (anode):
bubbles / gas;
Award mark for observation even if type of gas is incorrect (e.g. hydrogen).

Observation at negative electrode (cathode):
red/brown/copper/metal (deposit);
Allow mass increases / gets thicker/larger / OWTTE.

Change in colour (in any) of the solution:
solution loses blue colour/becomes paler;

Change in acidity (if any) of the solution:
becomes (more) acidic / pH decreases; ***[7]***

(iii) *Half-equation at positive electrode (anode)*:
$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$;
Allow e instead of e^{-}.

Half-equation at negative electrode (cathode):
$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$;
Allow e instead of e^{-}.
*Award **[1 max]** for M1 and M2 if correct equations are given but at wrong electrodes.*
Penalize $\rightleftharpoons$ once only in (b)(ii) and (b)(iii).

correct state symbols in **all** equations;
M3 can only be scored if the correct species are given in M1 and M2 (i.e. do not award ECF from M1 and M2 for incorrect species).

Observation at positive electrode (anode):
(slowly) dissolves / *OWTTE*;
Allow mass decreases / gets smaller/thinner / OWTTE;

Observation at negative electrode (cathode):
red/brown/copper/metal deposit;
Allow mass increases / gets thicker/larger / OWTTE.

Change in colour (if any) of the solution:
blue colour remains;
Allow no change or solution remains same colour.

Change in acidity (if any) of the solution:
solution does not become (more) acidic / no change / *OWTTE*; ***[7]***

LESSON 6.2

Balancing Redox Reactions

Answers for 2 multiple-choice and 1 structured questions

Part A - Multiple choice answers

6.2-M1 Nov 2022 · P1 · Q29 **D**

6.2-M3 May 2015 TZ2 · P1 · Q31 **D**

Part B - Structured questions markscheme

6.2-S3 May 2017 TZ2 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i) · also 1.2, 7.1

2.	**a**	**i**	$Sn^{2+}(aq) \rightarrow Sn^{4+}(aq) + 2e^-$ ✓	*Accept equilibrium sign.* *Accept* $Sn^{2+}(aq) - 2e \rightarrow Sn^{4+}(aq)$	**1**
2	**a**	**ii**	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 3Sn^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l) + 3Sn^{4+}(aq)$ ✓	*Accept equilibrium sign.*	**1**
2	**a**	**iii**	« $[K_2Cr_2O_7] = \frac{13.239\ g}{294.20\ g\ mol^{-1} \times 0.100\ dm^3}$ =» $0.450\ «mol\ dm^{-3}»$ ✓		**1**
2	**a**	**iv**	$n(Sn^{2+}) = «0.450\ mol\ dm^{-3} \times 0.01324\ dm^3 \times \frac{3\ mol}{1\ mol}$ =» $0.0179\ «mol»$ ✓ $«[Sn^{2+}] = \frac{0.0179\ mol}{0.0100\ dm^3}$ =» $1.79\ «mol\ dm^{-3}»$ ✓	*Award **[2]** for correct final answer.*	**2**
2	**b**	**i**	«$E^{\ominus}$ =» +0.85 «V» ✓	*Accept 0.85 V.*	**1**

LESSON 6.3

Reactivity

Answers for 2 multiple-choice and 2 structured questions

Part A - Multiple choice answers

6.3-M3 Nov 2023 TZ1 · P1 · Q29 **A**

6.3-M4 May 2021 TZ1 · P1 · Q28 **D**

Part B - Structured questions markscheme

6.3-S3 Nov 2015 · Paper 2 · Q7 parts (d)(i), (d)(ii), (e) · also 7.2

(d) (i) $Pb < Ni < Fe < Zn$;; ***[2]***

*Award **[2]** for the correct order.*
*Award **[1]** for $Zn > Fe > Ni > Pb$ as metals not listed in order of increasing reactivity.*
*Award **[1]** if one error in the order.*

(ii) Pb^{2+}/ lead(II) (ions); ***[1]***

Do not accept Pb/lead.

(e) *Dilute sodium chloride:*

$$H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- / 2OH^-(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^-;$$

C*oncentrated sodium chloride:*

$$Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-; \quad \textbf{[2]}$$

*Award **[1 max]** if equations are given the wrong way round.*
Accept e instead of e^-.
Ignore state symbols and equilibrium sign.
Accept correct half-equations balanced with different coefficients.

6.3-S1 May 2022 TZ1 · Paper 2 · Q2 parts (a), (b)(i), (b)(iii) · also 7.1, 7.2

2.	**a**		**Alternative 1** put Mg in Zn^{2+} (aq) ✓ Zn/«black» layer forms «on surface of Mg» ✓ **Alternative 2** place both metals in acid ✓ bubbles evolve more rapidly from Mg ***OR*** Mg dissolves faster ✓ **Alternative 3** construct a cell with Mg and Zn electrodes ✓ bulb lights up ***OR*** shows (+) voltage ***OR*** size/mass of Mg(s) decreases <<over time>> ***OR*** size/mass of Zn increases <<over time>> ✓	*Award **[1 max]** for "no reaction when Zn placed in Mg^{2+}(aq)".* *Accept "electrons flow from Mg to Zn".* *Accept Mg is negative electrode/anode* ***OR*** *Zn is positive electrode/cathode* *Accept other correct methods.*	**2**
2.	**b**	**i**	Cell potential: «(–0.45 V – (–2.37 V)» = «+»1.92 «V»✓		**1**

2.	**b**	**iii**	$2\ H_2O + 2\ e^- \rightarrow H_2 + 2\ OH^-$ ✓	*Accept equation with equilibrium arrows.*	**1**

LESSON 7.1

Primary & Secondary Cells

Answers for 9 multiple-choice and 5 structured questions

Part A - Multiple choice answers

7.1-M8	May 2023 TZ1 · P1 · Q31	A	7.1-M14	May 2022 TZ2 · P1 · Q30	A	7.1-M33	May 2015 TZ2 · P1 · Q32	C
7.1-M12	Nov 2022 · P1 · Q30	B	7.1-M18	May 2021 TZ2 · P1 · Q30	A	7.1-M41	May 2013 TZ1 · P1 · Q33	D
7.1-M29	May 2016 · P1 · Q31	D	7.1-M30	Nov 2015 · P1 · Q32	C	7.1-M56	May 2010 TZ2 · P1 · Q31	A

Part B - Structured questions markscheme

7.1-S1 May 2025 TZ3 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b)

6.	**(a)**	**(i)**	$Li \rightarrow Li^+ + e^-$ ***OR*** $LiC_6 \rightarrow C_6 + Li^+ + e^-$ ✓	*Do **not** accept 6C.* *Do **not** penalize equilibrium arrows.*	**1**
6.	**(a)**	**(ii)**	negative sign ***AND*** large value ✓ oxidation/reverse reaction is «highly» spontaneous/favourable ✓	*Accept Li is a good reductant/reducing agent for M2.*	**2**
6.	**(b)**		arrow from cathode to anode in the external circuit ✓	*Do **not** award mark if any arrows are in the electrolyte.*	**1**
				Accept “stores charge/power for night use”.	

7.1-S6 May 2024 TZ2 · Paper 3 · Q13 parts (a), (b)(i), (c)(i), (c)(ii), (c)(iii) · also 7.3

13.	**a**		energy density ***AND*** requires less storage space ***OR*** gases need **«**extra weight for**»** compression equipment ***OR*** **«**small volume**»** easier to transport/carry ***OR*** high specific energy with low energy density technologies are cost prohibitive ✓		**1**
13.	**b**	**i**	M_r ethylbenzene = 106.18 g mol^{-1} ✓ **«**$\Delta H_c = \frac{-4.135 \times 10^7 \text{ J kg}^{-1}}{1000 \text{ g kg}^{-1}} \times \frac{106.18 \text{ g mol}^{-1}}{1000 \text{ J kJ}^{-1}}$ **=»** –4391 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for missing negative sign in answer.* *Award **[2]** for -4380* **«**kJ mol^{-1}*» for M2 when 106 g mol^{-1} used for M1.*	**2**

13.	**c**	**i**	1 mol **«**of atoms only**»** 7 g for Li ***AND*** 207 g for Pb ***OR*** lower mass of Li for same number of atoms **«**oxidized**»** ***OR*** lower mass of Li per number of electrons **«**transferred**»** ✓ **«**voltage per cell**»** oxidation potential of Li(s) is +3.04 V ***AND*** Pb(s) is +0.13 V ***OR*** higher voltage for Li ✓	*Accept "higher charge density for Li" for M2.*	**2**
13.	**c**	**ii**	*Graphite:* $Li^+ + e^- \rightarrow Li(s)$ ***OR*** $6C(s, graphite) + Li^+ + e^- \rightarrow LiC_6(s)$ ✓ *Li CoO_2(s):* $LiCoO_2(s) \rightarrow Li^+ + CoO_2(s) + e^-$ ✓	*Award **[1 max]** for reversed half-equations given at the correct electrodes.*	**2**

13.	**c**	**iii**	sensitive to high temperature ***OR*** can overheat/explode/catch fire ***OR*** damaged if overcharged **«**so require circuit protection**»** ***OR*** ruined if completely discharged ***OR*** high cost ***OR*** ageing/loss of battery capacity/limited number of charge/discharge cycles ✓	*Apply List Principle*	**1**

7.1-S18 May 2018 TZ1 · Paper 2 · Q6 parts (a), (b), (c), (d), (e) · also 1.2

6.	**a**		salt bridge ✓ movement of ions ***OR*** balance charge ✓	*Do **not** accept "to complete circuit" unless ion movement is mentioned for M2.*	**2**
6.	**b**		*Positive electrode (cathode):* $Ag^+(aq) + e^- \rightarrow Ag(s)$ ✓ *Negative electrode (anode):* $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$ ✓	*Award **[1 max]** if correct equations given at wrong electrodes.*	**2**
6.	**c**		in external wire from left to right ✓		**1**
6.	**d**		«$E = +0.80\text{ V} - (-2.37\text{ V}) = +$» 3.17 «V» ✓		**1**
6.	**e**		«moles of silver $= \frac{0.10\text{ g}}{107.87\text{ g mol}^{-1}}$» moles of magnesium $= \frac{0.5 \times 0.10\text{ «g»}}{107.87\text{ «g mol}^{-1}\text{»}}$ ✓ «loss in mass of magnesium $= \frac{24.31\text{ g mol} \times 0.5 \times 0.10\text{ g}}{107.87\text{ g mol}^{-1}} =$ » 0.011 «g» ✓	*Award **[2]** for correct final answer.*	**2**

7.1-S21 May 2017 TZ1 · Paper 2 · Q3 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1, 6.2

3.	**a**		V_2O_5: +5 ✓ VO^{2+}: +4 ✓	*Do **not** penalize incorrect notation twice.*	**2**
3.	**b**	**i**	H_2SO_3 (aq) ***OR*** Pb (s) ✓		**1**
3.	**b**	**ii**	Zn (s) ✓		**1**
3.	**c**	**i**	VO^{2+} (aq) + V^{2+} (aq) + $2H^+$ (aq) → $2V^{3+}$ (aq) + H_2O (l) ✓	*Accept equilibrium sign.*	**1**

7.1-S29 Nov 2013 · Paper 2 · Q6 parts (a), (b)(i), (c), (d)(i), (d)(ii), (d)(iii), (e)(iii), (f)(i), (f)(ii) · also 3.1, 4.2, 6.1

6. (a) potential of the half-cell / reduction half-reaction under standard conditions measured relative to standard hydrogen electrode/SHE; ***[1]***

Allow instead of standard conditions, solute concentration of 1 mol dm^{-3} or 1 bar/1 atm (pressure) for gases.

(b) (i)

Initial oxidation number		Final oxidation number	Oxidized / reduced
IV/+4	**and**	III/+3;	reduced;

[2]

+ sign must be present. Do not award mark for incorrect notation 4, 4+, 3, 3+ etc.

Do not award M2 if inconsistent with M1.

(c) *KI (aq) is added to a solution containing $Ti^{3+}(aq)$ ions:*
no reaction;

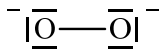
Zn (s) is added to a solution containing $TiO^{2+}(aq)$ and $H^{+}(aq)$ ions:
$Zn(s) + 2TiO^{2+}(aq) + 4H^{+}(aq) \rightarrow Zn^{2+}(aq) + 2Ti^{3+}(aq) + 2H_2O(l)$
correct reactants **and** products;
balanced equation; ***[3]***
Ignore state symbols.

(d) (i) $Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$;
Ignore state symbols.
Accept equilibrium arrow.

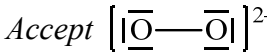
reduction; ***[2]***
Do not apply ECF.

(ii) $(+0.77 - (-0.06)) = (+)0.83(V)$; ***[1]***
Do not accept –0.83 V.

(iii) wire and salt bridge **both** have arrows from B to A; ***[1]***
Accept arrows above or below each provided it is obvious which they refer to.
Apply ECF from part (i).

(iii) 

e-pairs correct;
charges in correct positions; ***[2]***
Accept lines, or pairs of dots or crosses, for electron pairs.

Accept  .

(f) (i) C is sp hybridized **and** Si is sp^3 hybridized;
C–O bond in CO_2 has one σ-bond and one π-bond;
Si–O bond in SiO_2 has one σ-bond only; ***[3]***
*Award **[1 max]** for last two marking points for "C–O double bond and Si–O single bond".*

(ii) silicon-oxygen bonds will have a tetrahedral distribution;
xenon-oxygen bonds will have a square planar distribution;
xenon dioxide has **two** non-bonding/lone pairs of electrons; ***[3]***
Award any of the above marks if clearly indicated in suitable diagrams.

LESSON 7.2

Electrolytic Cells

Answers for 4 multiple-choice and 3 structured questions

Part A - Multiple choice answers

7.2-M6	May 2023 TZ2 · P1 · Q31	**A**	**7.2-M9**	May 2021 TZ1 · P1 · Q31	**C**
7.2-M8	Nov 2021 · P1 · Q31	**A**	**7.2-M26**	Nov 2013 · P1 · Q34	**D**

Part B - Structured questions markscheme

7.2-S19 May 2014 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(v), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i), (c)(ii), (d)(ii), (d)(iii) · also 6.2, 6.3, 7.1, 7.3

6. (a) (i) $C_4H_9OH(l) \rightarrow C_4H_8O(l) + 2H^+(aq) + 2e^-$; ***[1]***

Ignore state symbols.

(ii) $3C_4H_9OH(l) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \rightarrow 3C_4H_8O(l) + 2Cr^{3+}(aq) + 7H_2O(l)$; ***[1]***

Ignore state symbols.

(v) $C_4H_9OH + 6O_2 \rightarrow 4CO_2 + 5H_2O$ / $(CH_3)_3COH + 6O_2 \rightarrow 4CO_2 + 5H_2O$

correct reactants and products;

correct balancing; ***[2]***

(b) (i) Z < W < X < Y;

Accept Y > X > W > Z. ***[1]***

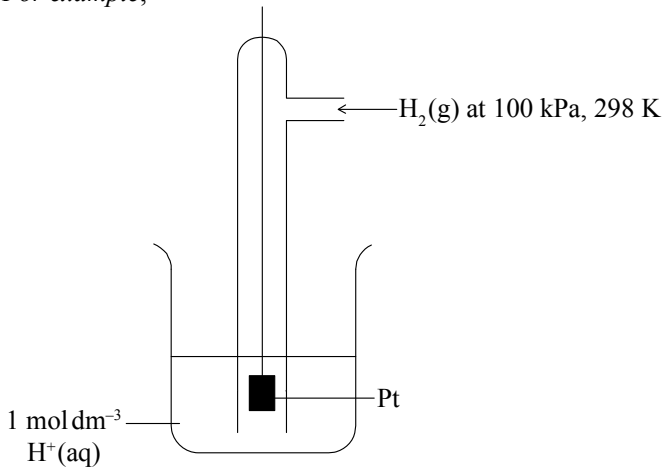
(ii) $X(s) + Z^{2+}(aq) \rightarrow X^{2+}(aq) + Z(s)$; ***[1]***

Ignore state symbols.

Accept X(s) + ZCl_2(aq) → XCl_2(aq) + Z(s).

(iii) H_2(g)/hydrogen; ***[1]***

(iv) diagram showing gas, solution and solid electrode;
For example,



*This diagram scores **[3]**.*
$1\,\text{mol}\,\text{dm}^{-3}\ H^+(aq)$ **and** Pt;
Allow 1 mol L^{-1} or 1 M.
Allow 1 mol dm^{-3} HCl (aq) or other source of 1 mol dm^{-3} $H^+(aq)$ ions.

100 kPa/10^5 Pa/1 bar ($H_2(g)$ pressure) **and** 298 K / 25 °C; ***[3]***
Ignore state symbols throughout.
Allow 1.01×10^5 Pa/1 atm.

(c) (i) *Positive electrode (anode):*
$I^-(aq) \rightarrow \frac{1}{2}I_2(aq) + e^-$;
Accept correct equation involving 2 mols of I^-.

Negative electrode (cathode):
$H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq)$ / $H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$ /
$H_3O^+(aq) + e^- \rightarrow H_2O(l) + \frac{1}{2}H_2(g)$; ***[2]***
*Award **[1 max]** if correct equations are given at the wrong electrodes.*
Ignore state symbols.
Allow e instead of e^-.
Penalize equilibrium sign once only.
Accept correct equation involving 2 mols of H^+.

(ii) aluminium will be oxidized (instead of I^-) at positive electrode (anode);
aluminium is a reactive metal / oxidation of aluminium has a positive $E^\ominus$ /
aluminium is higher on the reactivity series than I^- / *OWTTE*; ***[2]***

(ii) blue colour persists in second cell **and** fades in third cell;
pH does not change in second cell **and** decreases in third cell; ***[2]***
*Award **[1 max]** if both colour and pH are correctly stated for one only of either second or third cell.*

(iii) *Colour:*
positive Cu electrode (anode) is oxidized to maintain colour in second cell / $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$;

pH:
in third cell, H^{+} ions are produced as water is oxidized at positive electrode (anode) / $H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^{+}(aq) + 2e^{-}$ / solution becomes acidic as hydroxide ions are oxidized at positive electrode (anode) / $2OH^{-}(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^{-}$; ***[2]***
Ignore state symbols.

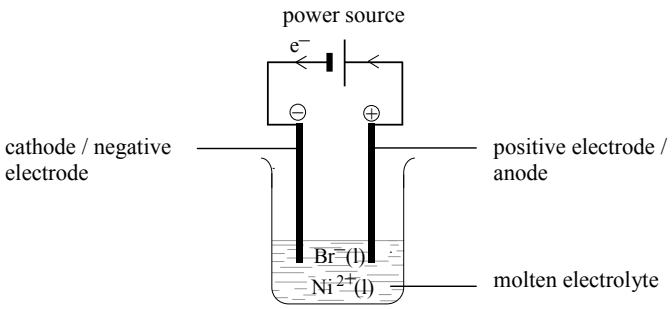
7.2-S7 Nov 2019 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (c)(iv), (e)(i), (e)(ii) · also 1.2, 2.1, 7.1, 9.2

6.	**a**	**i**	[Ar] $3d^{10}$ ***OR*** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ ✓		**1**
6.	**a**	**ii**	$\Delta H^{\ominus} = \sum \Delta H^{\ominus}_f$ (products) – $\sum \Delta H^{\ominus}_f$ (reactants) ✓ $\Delta H^{\ominus}$ = 2(–241.8 «kJ mol^{-1}») – 4(–92.3 «kJ mol^{-1}») = –114.4 «kJ» ✓	*Award **[2]** for correct final answer.*	**2**

6.	**b**		mass of H_2O = «18.360 g – 17.917 g =» 0.443 «g» ***AND*** mass of $CuCl_2$ = «17.917 g – 16.221 g =» 1.696 «g» ✓ moles of $H_2O = «\frac{0.443\,g}{18.02\,g\,mol^{-1}} =»$ 0.0246 «mol» ***OR*** moles of $CuCl_2 = «\frac{1.696\,g}{134.45\,g\,mol^{-1}} =»$ 0.0126 «mol» ✓ «water : copper(II) chloride = 1.95 : 1» «x =» 2 ✓	*Award **[3]** for correct final answer.* *Accept «x =» 1.95.*	**3**
6.	**c**	**i**	*Wires:* «delocalized» electrons «flow» ✓ *Electrolyte:* «mobile» ions «flow» ✓		**2**
6.	**c**	**ii**	$2Cl^- \rightarrow Cl_2(g) + 2e^-$ ***OR*** $Cl^- \rightarrow \frac{1}{2}Cl_2(g) + e^-$ ✓	*Accept e for* e^-.	**1**
6.	**c**	**iii**	«electrode» 3 ***AND*** oxygen/O_2 ✓	*Accept chlorine/Cl_2.*	**1**
6.	**c**	**iv**	$2H_2O(l) \rightarrow 4H^+(aq) + O_2(g) + 4e^-$ ✓	*Accept $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$.* *Accept $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$*	**1**
6.	**e**	**i**	$E^\ominus$ = «+0.52 – 0.15 = +» 0.37 «V» ✓		**1**
6.	**e**	**ii**	spontaneous ***AND*** $E^\ominus$ positive ✓		**1**

7.2-S26 May 2011 TZ1 · Paper 2 · Q9 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c), (e)(i), (e)(ii) · also 6.1, 6.2, 7.1

9. (a) (i)



labelled polarities of positive **and** negative electrodes;
Electrodes can be labelled positive or negative or + and - signs can be used.

direction of electron flow;
e^- does not have to be labelled but arrow essential.

power source **and** molten electrolyte/Ni^{2+} (l) **and** Br^- (l)/$NiBr_2$ (l);
State symbol necessary for M3 unless molten electrolyte stated.
Power source does not need to be labelled if correct symbol used (i.e. short line and long line).

Cathode/negative electrode equation:
$Ni^{2+} + 2e^- \rightarrow Ni$;
Anode/positive electrode equation:
$2Br^- \rightarrow Br_2 + 2e^-$; ***[5]***
Accept balanced half-equation with one e^-.
*Award **[1 max]** for M4 and M5 if electrodes are not identified or if equations are given wrong way round or incorrectly labelled.*
Penalize $\rightleftharpoons$ once only in Q.9.
Allow e instead of e^-.
Ignore state symbols for M4 and M5.

(ii) *Dilute sodium chloride:*
$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$ / $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$;

Concentrated sodium chloride:
$2Cl^- \rightarrow Cl_2 + 2e^-$; ***[2]***
Accept alternative balanced half-equations with correct number of electrons.
*Award **[1 max]** if equations are given the wrong way round.*
*Award **[2]** if correct equations are written in order with dilute sodium chloride first and concentrated sodium chloride second but processes not stated explicitly.*
Penalize $\rightleftharpoons$ once only in Q.9.
Allow e instead of e^-.
Ignore state symbols

(b) (i) $Sn + Cu^{2+} \rightarrow Sn^{2+} + Cu$; ***[1]***
Ignore state symbols.
Penalize $\rightleftharpoons$ *once only in Q.9.*

(ii) (0.34 – –0.14) = (+) 0.48 V; ***[1]***

(iii) 1.0 mol dm^{-3} solutions **and** 25 °C /298 K; ***[1]***

(c) Cd^{2+} is a stronger oxidizing agent than H_2O **and** will be displaced to produce Cd / *OWTTE*;
Cr^{2+} is a weaker oxidizing agent than H_2O **and** H_2 will displace in preference to Cr / *OWTTE*; ***[2]***
*Award **[1 max]** for stating Cd^{2+} stronger oxidizing agent than H_2O **and** Cr^{2+} weaker oxidizing agent than H_2O / OWTTE.*

(e) (i) *NH_3:* –3;
NO: +2;
N_2: 0; ***[3]***
Penalize incorrect notation such as 3–, III, 2+, 2, II once only.

(ii) *Oxidation:*
$2NH_3 \rightarrow N_2 + 6H^+ + 6e^-$;
Reduction:
$2NO + 4H^+ + 4e^- \rightarrow N_2 + 2H_2O$;
*Award **[1 max]** for M1 and M2 if redox processes are not identified or if equations are given wrong way round.*
Penalize $\rightleftharpoons$ *once only in Q.9.*
Allow e instead of e^-.
Ignore state symbols.

Oxidizing agent: NO; ***[3]***
Allow either formula or name.

LESSON 7.3

Energy From Fuels

Answers for 2 multiple-choice and 3 structured questions

Part A - Multiple choice answers

7.3-M3 May 2017 TZ2 · P1 · Q1 **C**

7.3-M4 May 2015 TZ2 · P1 · Q2 **C**

Part B - Structured questions markscheme

7.3-S1 May 2025 TZ3 · Paper 2 · Q9 parts (b)(i), (b)(ii), (c) · also 8.2, 9.3

9.	**(b)**	**(i)**	*Any two:* «ethanol is» renewable / sustainable resource ✓ «ethanol has» low/zero carbon footprint / produces less CO_2 ✓ less sulfur dioxide «than fossil fuels» ***OR*** less acid rain «than fossil fuels» ✓ less incomplete combustion «than fossil fuels» ***OR*** less carbon monoxide/soot «than fossil fuels» ✓	*Accept "ethanol is biodegradable /less toxic than gasoline".* *Do **not** accept just "less harmful".*	**2 max**
9.	**(b)**	**(ii)**	«bond breaking» 1 C-C + 5 C-H + 1 C-O + 1 O-H + 3 O=O ***OR*** 346 + 5(414) + 358 + 463 + 3(498) ***OR*** 4731 «kJ» ✓ «bond forming» 4 C=O + 6 O-H ***OR*** 4(804) + 6(463) ***OR*** 5994 «kJ» ✓ «ΔH = 4731 – 5994 =» -1263 «kJ mol^{-1}» ✓	*Award **[3]** for correct final answer.*	**3**
9.	**(c)**		«Q = (4.00 / 46.08) x 1367 =» 119 kJ / 119000 J ✓ «ΔT = Q / mc = 119000 / (500.0 x 4.18) =» 56.9 «K» ✓	*Accept 56.8 «K».* *Allow use of ΔH_c calculated in (b)(ii) without penalty. Answer is ΔT = 52.5 / 52.6 «K».*	**2**

7.3-S3 May 2024 TZ1 · Paper 3 · Q13 parts (a), (b), (c), (f)

13.	**a**		$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ ✓		**1**
13.	**b**		amount of CH_4 in 1.00 g clathrate «$= 4 \times \frac{1\text{ g}}{478.7\text{ g mol}^{-1}}$» = 0.008356 «mol» ✓ specific energy « = 0.008356 mol g^{-1} × 891 kJ mol^{-1}» = 7.45 «kJ g^{-1}» ✓	*Award **[2]** for correct final answer.*	**2**
13.	**c**		*Lower specific energy:* not all methane clathrate is methane ***OR*** part of methane clathrate does not burn ✓ *Higher energy density:* methane is gas ***AND*** methane clathrate is solid ***OR*** methane has much lower density «than methane clathrate» ***OR*** methane clathrate occupies a much smaller volume than «same mass of» methane ✓	*For M1 accept reference to water molecules in methane clathrate as indicating it is not all methane.*	**2**

13.	**f**		*Support:* overall increase in both the carbon dioxide concentration and the temperature «anomaly» ✓ *Refute:* CO_2 increases between ~1940 and ~1970 without corresponding increase in temperature ***OR*** global temperatures rose between ~1935 and ~1945 despite [CO_2] remaining relatively constant ***OR*** increases in global temperature between ~1900 and ~1940 and between ~1970 and ~2010 are similar, but increase in CO_2 levels much higher in latter period ✓	*For M1, accept there is a good/close correlation «in their shapes».* *For M2, accept the connection may just be coincidental «not causative».* *Do **not** accept comments about fluctuations.* *Accept years that are approximately the same as these listed.*	**2**

7.3-S19 May 2019 TZ1 · Paper 3 · Q17 parts (b)(i), (b)(ii), (b)(iii), (c)

17.	**b**	**i**	Electrode A: $C_2H_6O\,(aq) + 3H_2O\,(l) \rightarrow 12H^+\,(aq) + 12e^- + 2CO_2\,(g)$ ✓ Electrode B: $3O_2\,(g) + 12H^+\,(aq) + 12e^- \rightarrow 6H_2O\,(l)$ ✓	*Accept balanced equations with integer or fractional coefficients.* *Penalize equilibrium arrows once only.*	**2**
17.	**b**	**ii**	*Name:* PEM/proton-exchange membrane/polymer exchange membrane/polymer electrolyte membrane ✓ *Function:* allows the passage of protons/H^+ ions **«**from anode to cathode but not electrons or molecules**»** ✓		**2**
17.	**b**	**iii**	*Any one of:* water is a reactant/allows the cell to operate at a higher concentration of protons/H^+ ions ***OR*** water is a stronger electrolyte and thus produces higher electric current ✓ less dangerous/flammable ✓		**1**
17.	**c**		use of **«**farm**»** land **«**for production**»** ***OR*** deforestation **«**for crop production for fuel**»** ***OR*** can release more NO_x **«**than normal fuel on combustion**»** ✓	*Ignore any reference to cost.*	**1**

LESSON 8.1

Thermochemical Equations

Answers for 3 multiple-choice and 1 structured questions

Part A - Multiple choice answers

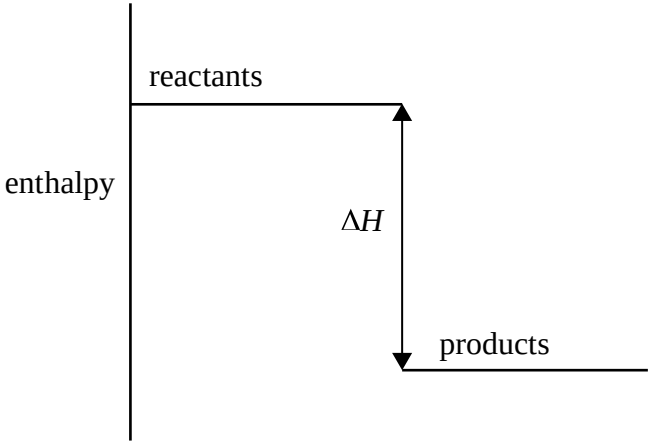
8.1-M4	Nov 2023 TZ1 · P1 · Q14	**D**	**8.1-M5**	Nov 2023 TZ1 · P1 · Q16	**A**	**8.1-M9**	Nov 2020 · P1 · Q15	**B**

Part B - Structured questions markscheme

8.1-S8 May 2012 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (c) · also 8.2

8. (a) (i) enthalpy on *y*-axis;
Do not accept energy.

reactants higher than products;
ΔH labelled; ***[3]***



Accept appropriate formulas for reactants and products.
Arrow heads not needed.
57.6 is acceptable as an alternative to ΔH .

(ii) products are more stable as they are at a lower enthalpy level / energy has been given off by the reactants / reaction is exothermic / *OWTTE*; ***[1]***

(iii) $n(\text{NaOH}) = 0.125\ \text{mol}$;
change in heat energy = $(-57.6 \times 0.125) = -7.20$ (kJ) / heat released $= (57.6 \times 0.125) = 7.20\ (\text{kJ})$; ***[2]***

(c) $q = (mc\Delta T =) 100.0 \times 4.18 \times 3.50 / 1463\ \text{J} / 1460\ \text{J}$;

$n(\text{NH}_4\text{Cl}) = \dfrac{5.35}{53.5} / 0.100\ \text{mol}$;

ΔH = +14.6 / 14.6 (kJ mol^{-1}); ***[3]***

Accept $q = 105.35 \times 4.18 \times 3.50 / 1541\ J$.

Accept ΔH = *+15.4 / 15.4 (kJ* mol^{-1}*).*

LESSON 8.2

Calorimetry

Answers for 4 multiple-choice and 2 structured questions

Part A - Multiple choice answers

8.2-M1	May 2025 TZ1 · P1A · Q23	**C**	**8.2-M10**	May 2022 TZ2 · P1 · Q13	**A**
8.2-M8	May 2023 TZ2 · P1 · Q15	**C**	**8.2-M12**	May 2021 TZ1 · P1 · Q38	**A**

Part B - Structured questions markscheme

8.2-S17 May 2011 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 1.2, 9.1

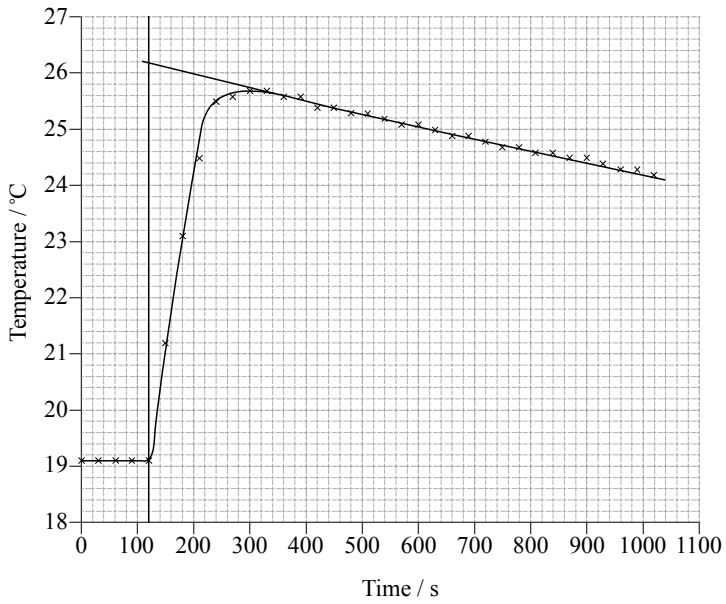
(a) (i) $\text{amount} = \frac{3.99}{159.61} = 0.0250\,(\text{mol});$ ***[1]***

(ii) $26.1\,(^{\circ}\text{C});$

Accept answers between 26.0 and 26.2 (°C).

$\text{temperature rise} = 26.1 - 19.1 = 7.0\,(^{\circ}\text{C});$ ***[2]***

Accept answers between 6.9 °C and 7.1 (°C).
*Award **[2]** for the correct final answer.*
No ECF if both initial and final temperatures incorrect.



(iii) heat change $= \frac{50.0}{1000} \times 4.18 \times 7.0\,(\text{kJ}) / 50.0 \times 4.18 \times 7.0\,(\text{J})$;

Accept 53.99 instead of 50.0 for mass.

$= 1.5\,(\text{kJ})$; ***[2]***

Allow 1.6 (kJ) if mass of 53.99 is used.
Ignore sign.

(iv) $\Delta H_1 = -\frac{1.5}{0.0250} = -60\,(\text{kJ mol}^{-1})$; ***[1]***

Value must be negative to award mark.
Accept answers in range –58.0 to –60.0.
Allow –63 (kJ mol^{-1}) if 53.99 g is used in (iii).

(b) (i) (amount of $CuSO_4 \cdot 5H_2O = \frac{6.24}{249.71} =) 0.0250\,(\text{mol})$;

(amount of H_2O in 0.0250 mol of $CuSO_4 \cdot 5H_2O = 5 \times 0.0250 =) 0.125\,(\text{mol})$; ***[2]***

(ii) $(50.0 \times 4.18 \times 1.10 =) 230\,(\text{J})$;

$\left(\frac{229.9}{(1000 \times 0.0250)} =\right) + 9.20\,(\text{kJ})$; ***[2]***

Accept mass of 47.75 or 53.99 instead of 50.00 giving answers of +.8.78 or +9.9.
Do not penalize missing + sign but penalize – sign unless charge already penalized in (a) (iv).

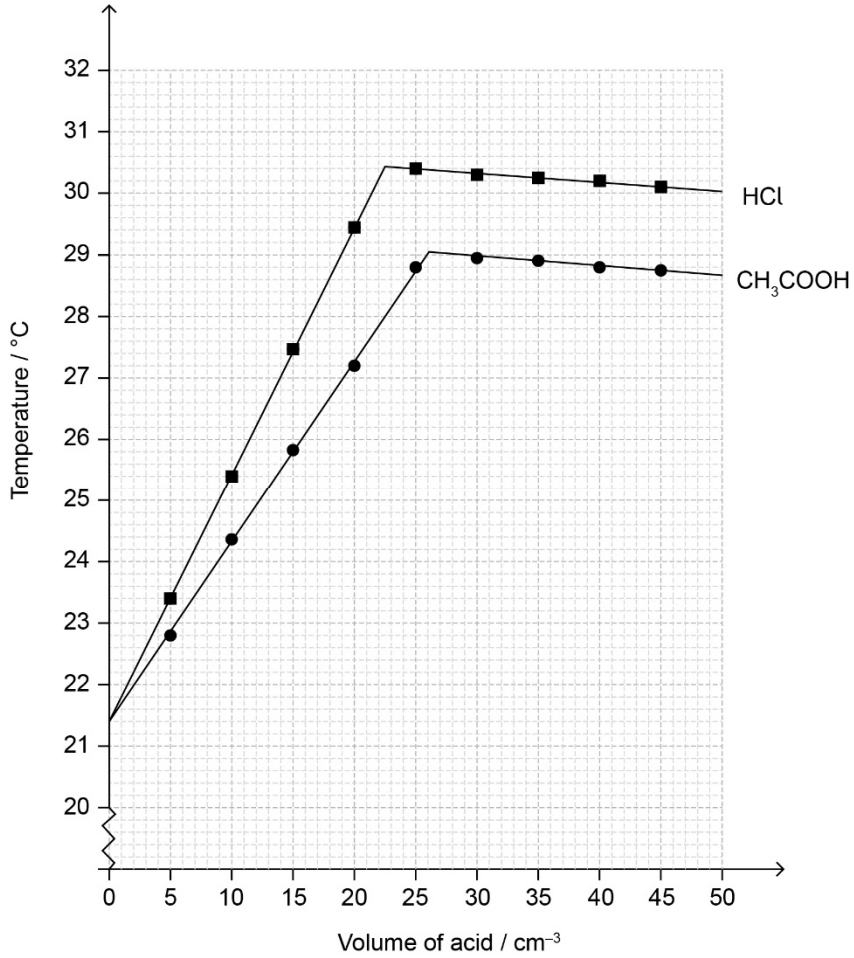
(iii) $(\Delta H_x = \Delta H_1 - \Delta H_2 = -58.4 - (+9.20) =) -67.6\,(\text{kJ mol}^{-1})$; ***[1]***

(c) (i) $\frac{[-78.0 - (-67.6)]}{-78.0} \times 100 = 13.3\%$; ***[1]***

If 70.0 kJ mol^{-1} is used accept 10.3%.

(ii) the anhydrous copper(II) sulfate had already absorbed some water from the air / *OWTTE*;
the value would be less exothermic/less negative than expected as the temperature increase would be lower / less heat will be evolved when the anhydrous salt is dissolved in water / *OWTTE*; ***[2]***
Do not accept less <u>without</u> a reason.

8.2-S8 Nov 2017 · Paper 2 · Q1 parts (a), (b), (c), (d)(i), (d)(ii) · also 1.2

1.	**a**		 21.4 °C ✓	*Accept values in the range of 21.2 to 21.6* °C. *Accept two different values for the two solutions from within range.*	**1**

1.	**b**		*HCl:* 30.4 **«**°C**»** ✓ *CH_3COOH:* 29.0 **«**°C**»** ✓	*Accept range 30.2 to 30.6* °C. *Accept range 28.8 to 29.2* °C.	**2**
1.	**c**		***ALTERNATIVE 1*** **«**volume CH_3COOH **=»** 26.0 **«**cm^3**»** ✓ **«**$[CH_3COOH] = 0.995 \text{ mol dm}^{-3} \times \frac{50.0\,\text{cm}^3}{26.0\,\text{cm}^3}$ **=»** 1.91 **«**mol dm^{-3}**»** ✓ ***ALTERNATIVE 2*** **«***n*(NaOH) = 0.995 mol dm^{-3} × 0.0500 dm^3 **=»** 0.04975 **«**mol**»** ✓ **«**$[CH_3COOH] = \frac{0.04975}{0.0260}\text{dm}^3$ **=»** 1.91 **«**mol dm^{-3}**»** ✓	*Accept values of volume in range 25.5 to 26.5 cm^3.* *Award **[2]** for correct final answer.*	**2**
1.	**d**	**i**	**«**total volume $= 50.0 + 26.0$ **=»** 76.0 cm^3 ***AND*** **«**temperature change $29.0 - 21.4$ **=»** 7.6 **«**°C**»** ✓ **«***q* = 0.0760 kg × 4.18 kJ kg^{-1} K^{-1} × 7.6 K **=»** 2.4 **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**
1.	**d**	**ii**	**«***n*(NaOH) = 0.995 mol dm^{-3} × 0.0500 dm^3 **=»** 0.04975 **«**mol**»** ***OR*** **«***n*(CH_3COOH) = 1.91 mol dm^{-3} × 0.0260 dm^3 **=»** 0.04966 **«**mol**»** ✓ **«** $\Delta H = -\frac{2.4\,\text{kJ}}{0.04975\,\text{mol}}$ **=»** −48 / −49 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *Negative sign is required for M2.*	**2**

LESSON 9.1

Hess's Law

Answers for 3 multiple-choice and 1 structured questions

Part A - Multiple choice answers

9.1-M1 Nov 2024 · P1 · Q16 **D**

9.1-M2 Nov 2023 TZ1 · P1 · Q15 **C**

9.1-M6 May 2018 TZ1 · P1 · Q14 **A**

Part B - Structured questions markscheme

9.1-S2 May 2017 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii) · also 8.2

9.	**a**	**i**	$\Delta H = 177.0 - \frac{189.2}{2} - 285.5$ «kJ» ✓ «ΔH =» – 203.1 «kJ» ✓	*Accept other methods for correct manipulation of the three equations.* *Award **[2]** for correct final answer.*	**2**
	a	**ii**	203.1 «kJ» = 0.850 «kg» × 4.18 «kJ kg^{-1} K^{-1}» × ΔT «K» ***OR*** «ΔT =» 57.2 «K» ✓ «T_{final} = (57.2 + 21.8)°C =» 79.0«°C» / 352.0 «K»✓ *If 200.0 kJ was used:* 200.0 «kJ» = 0.850 «kg» × 4.18 «kJ kg^{-1} K^{-1}» × ΔT «K» ***OR*** «ΔT =» 56.3«K» ✓ «T_{final} = (56.3 + 21.8)°C =» 78.1 «°C» / 351.1 «K»✓	*Award **[2]** for correct final answer.* *Accept two, three or four significant figures.* *Unit, if specified, must be consistent with value stated.*	**2**

LESSON 9.2

Enthalpies of Formation & Combustion

Answers for 2 multiple-choice and 1 structured questions

Part A - Multiple choice answers

9.2-M11 Nov 2017 · P1 · Q16 **B**

9.2-M13 May 2016 · P1 · Q16 **A**

Part B - Structured questions markscheme

9.2-S2 Nov 2023 TZ1 · Paper 2 · Q4 parts (a)(i), (b) · also 3.1

4	**a**	**i**	«Σ $\Delta H^{\ominus}_f$ (reactants) =» +88.7 + 2(-241.8) / -394.9 «kJ mol^{-1}» ***AND*** «Σ $\Delta H^{\ominus}_f$ (products) =» -393.5 + 2(-20.6) / -434.7 «kJ mol^{-1}» ✓ $\Delta H^{\ominus}$ «= -434.7 – (-394.9)» = -39.8 «kJ mol^{-1}» ✓	*Award **[2]** for correct final answer.* *Award **[1]** for +48.2 «kJ mol^{-1}» (obtained using -285.8 kJ mol^{-1} for $H_2O(l)$).*	**2**
4	**b**		*Molecular geometry CS_2:* linear ***AND*** *Molecular geometry H_2S:* bent/V-shaped ✓ *Reason for difference:* «central atom in» H_2S has «two» lone/non-bonding «electron» pairs ***OR*** CS_2 has two ***AND*** H_2S has four electron domains/negative charge centres «around central atom» ✓	*Do **not** accept diagrams for M1 or M2.* *Accept central atom sp hybridized in CS_2 **AND** sp^3 hybridized in H_2S for M2.*	**2**

LESSON 9.3

Bond Enthalpies

Answers for 2 multiple-choice and 3 structured questions

Part A - Multiple choice answers

9.3-M5 May 2016 · P1 · Q17 **B**

9.3-M6 May 2015 TZ2 · P1 · Q15 **B**

Part B - Structured questions markscheme

9.3-S1 May 2025 TZ1 · Paper 2 · Q8 parts (c)(v), (f)(i), (f)(ii), (f)(iii) · also 5.2, 9.2

8.	**(c)**	**(v)**	stronger forces of attraction with increasing chain length / larger electron cloud /molar mass ✓	*forces for M1.* *Accept "more electrons" for "larger electron cloud" in M2.* *Accept increased surface area.*	**2**
8.	**(f)**	**(i)**	*ΔH* = «3196 – 3237 = » −41 «kJ » ✓ **ALTERNATIVE 2:** Bonds broken: 1 C=C + 1 O-H / 614 + 463 / 1077 «kJ» ✓ Bonds formed: 1 C-C + 1 C-O + 1 C-H / 346 + 358 + 414 / 1118 «kJ» ✓ *ΔH* = «1077 – 1118 = » −41 «kJ» ✓	*answer.* *Award [2 max] for +41* «kJ».	**3**
8.	**(f)**	**(ii)**	« $\Delta H^{\ominus} = \Sigma(\Delta H_f^{\ominus}\text{products}) - \Sigma(\Delta H_f^{\ominus}\text{reactants})$ » = −235 – (+52 + (−242)) ✓ = −45 «kJ» ✓	*Award [2] for correct final answer.* *Award **[1 max]** for +45* «kJ»	**2**
8.	**(f)**	**(iii)**	bond enthalpies are average values ***OR*** bond enthalpies vary «slightly» among compounds ✓		**1**

9.3-S8 May 2018 TZ1 · Paper 2 · Q3 parts (c)(i), (c)(ii), (c)(iii), (d) · also 4.1, 9.2

3.	**c**	**i**	$\Delta H^{\ominus}$ = bonds broken – bonds formed ✓ «$\Delta H^{\ominus}$ = 3(C≡C) – 6(C⎓C)$_{benzene}$ / 3 × 839 – 6 × 507 / 2517 – 3042 =» –525 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for "+525 «kJ»".* *Award **[1 max]** for:* *«$\Delta H^{\ominus}$ = 3(C≡C) – 3(C–C) – 3(C=C) / 3 × 839 – 3 × 346 – 3 × 614 / 2517 – 2880 =» –363 «kJ».*	**2**
3.	**c**	**ii**	$\Delta H^{\ominus} = \Sigma\Delta H_f$ (products) – $\Sigma\Delta H_f$ (reactants) ✓ «$\Delta H^{\ominus}$ = 49 kJ – 3 × 228 kJ =» –635 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for "+635 «kJ»".*	**2**
3.	**c**	**iii**	ΔH_f values are specific to the compound ***OR*** bond enthalpy values are averages «from many different compounds» ✓ condensation from gas to liquid is exothermic ✓	*Accept "benzene is in two different states «one liquid the other gas»" for M2.*	**2**
3.	**d**		equal C–C bond «lengths/strengths» ***OR*** regular hexagon ***OR*** «all» C–C have bond order of 1.5 ***OR*** «all» C–C intermediate between single and double bonds ✓	*Accept "all C–C–C bond angles are equal".*	**1**

9.3-S13 Nov 2012 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (c)(ii), (c)(iii) · also 3.1, 4.1, 4.2, 9.2

5. (a) energy needed to break (1 mol of) a bond in a gaseous molecule/state/phase;
average calculated from a range of similar compounds / *OWTTE*; ***[2]***
Do not accept similar bonds instead of similar compounds.
M2 can be scored independently.

(b) (i) *Bonds breaking*:
$4 \times (N{=}O) + 4 \times (N{-}N) + 8 \times (C{-}H) + 4 \times (N{-}O) + 8 \times (C{-}N)$ /
$= (4)(607) + (4)(158) + (8)(413) + (4)(201) + (8)(286)$ /
$= 9456 \text{ (kJ mol}^{-1}\text{)}$;

Bonds forming:
$4 \times (N{\equiv}N) + 4 \times (C{\equiv}O) + 8 \times (O{-}H)$ /
$= (4)(945) + (4)(1072) + (8)(464) / = 11780 \text{ (kJ mol}^{-1}\text{)}$;

Enthalpy change:
$(9456 - 11780 =) - 2324 \text{ (kJ mol}^{-1}\text{)}$;
M3 is scored from M1-M2.
*Award **[2 max]** for (+)2324 (kJ mol^{-1}).*
*Award **[3]** for –2324 (kJ mol^{-1}) without working.*

$\left(10.0 \times \frac{-2324}{296.2} = \right) - 78.5 \text{ (kJ)}$; ***[4]***
M4 is scored from 10.0/296.2 × M3.
Allow answers of –78.2, –78 and –79 (kJ), but negative sign must be included.
*Award **[4]** for correct final answer.*

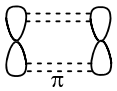
(ii) NH_4^+/ammonium / $N_2H_5^+$/hydrazinium / $CH_3NH_3^+$/methylammonium/ methanaminium / $H_2NO_3^+$ /nitrooxonium; ***[1]***

(iii) *Sigma bonds*:



result from head-on/end-on overlap of orbitals / *OWTTE*;
Allow symmetric (orbital) with respect to same plane / OWTTE.

Pi bonds:



result from sideways overlap of orbitals / *OWTTE*;
Allow antisymmetric (orbitals) with respect to (defining) plane (containing at least one atom) / OWTTE.

suitable diagrams showing σ **and** π bonds after formation; ***[3]***

*Award **[1 max]** for correct diagram without description given or **[2 max]** for*

(iv) *σ bonds*: 28;
π bonds: 4; ***[2]***

(v) mixing/combining/merging of atomic orbitals to form new/molecular orbitals (for bonding);
A: sp^2;
B: sp^3; ***[3]***
*Award **[1 max]** for M2 and M3 for sp^2 **and** sp^3 if **A** and **B** are not identified explicitly but do not award M2 and M3 if sp^3 is given for **A** and sp^2 for **B**.*

(ii) heat/enthalpy <u>change</u> when 1 mol of a compound is formed from its <u>elements</u>;
in their standard states / at 100 kPa/10^5 Pa/1 bar (pressure); ***[2]***
Allow 1.01×10^5 Pa / 101 kPa / 1 atm.
Allow under standard conditions or standard ambient temperature and pressure.
Temperatures not required in definition, allow if quoted (e.g. 298 K/25 °C – most common) but correct pressure value must be stated if given.
Only award M2 for a correct M1 or a near miss at M1.

(iii) $\Delta H^{\ominus} = (-485 + 111 + 239 =) -135$ (kJ mol^{-1}); ***[1]***

LESSON 9.4

Born-Haber Cycles

Answers for 2 multiple-choice and 2 structured questions

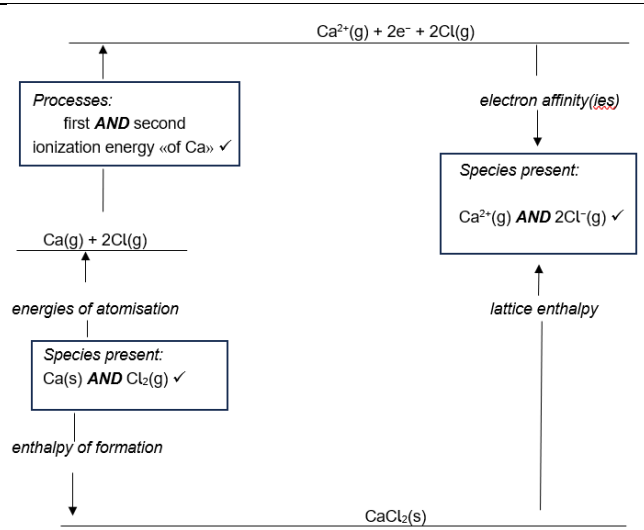
Part A - Multiple choice answers

9.4-M6 Nov 2021 · P1 · Q16 **A**

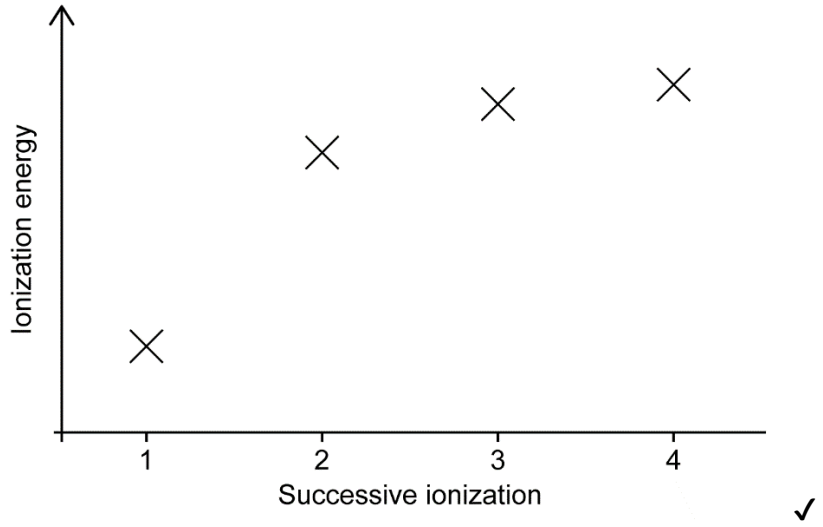
9.4-M9 May 2019 TZ1 · P1 · Q17 **A**

Part B - Structured questions markscheme

9.4-S1 May 2025 TZ1 · Paper 2 · Q4 parts (c)(i), (c)(ii)

4.	**(c)**	**(i)**		*Do not accept ionization energy for Processes.* *Accept ionization energies for Processes.*	**3**
4.	**(c)**	**(ii)**	Ca^{2+} has larger ionic radius «than Mg^{2+}» ***OR*** greater distance between ion centres «in $CaCl_2$ » ✓	*Must be clear reference to ionic for M1*	**1**

9.4-S3 May 2019 TZ1 · Paper 2 · Q3 parts (a), (d)(i), (d)(ii), (d)(iii), (f), (g)(i), (g)(ii), (h), (k) · also 1.2, 2.1, 4.1, 6.1, 9.2, 9.3

3.	**a**		 ✓	*Accept curve showing general trend.* *Award mark only if the energy difference between the first two points is larger than that between points 2/3 and 3/4.*	**1**

3.	**d**	**i**	$\frac{1}{2}\ O_2\,(g) \rightarrow O^{2-}\,(g)$ **«**$\Delta H_{atomisation}$ (O) + 1st EA + 2nd EA = 249 kJ mol^{-1} − 141 kJ mol^{-1} + 753 kJ mol^{-1} **=» «+»**861 **«**kJ mol^{-1}**»** ✓ Na (s) → Na^+ (g) **«**$\Delta H_{atomisation}$ (Na) + 1st IE = 107 kJ mol^{-1} + 496 kJ mol^{-1} **=» «+»**603 **«**kJ mol^{-1}**»** ✓		**2**
3.	**d**	**ii**	lattice enthalpy = 861 **«**kJ mol^{-1}**»** + 2 × 603 **«**kJ mol^{-1}**»** −(−414 **«**kJ mol^{-1}**»**) ✓ **«**= +**»** 2481 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *If given values are used:* *M1: lattice enthalpy = 850 **«**kJ mol^{-1}**»** + 2 × 600 **«**kJ mol^{-1}**»** −(−414 **«**kJ mol^{-1}**»**)* *M2: **«**= +**»** 2464 **«**kJ mol^{-1}**»***	**2**
3.	**d**	**iii**	K^+ ion is larger than Na^+ ***OR*** smaller attractive force because of greater distance between ion «centres» ✓		**1**

3.	**f**		n(Na_2O_2) theoretical yield **«**= $\frac{5.00\text{ g}}{61.98\text{ g mol}^{-1}}$ **»** = 0.0807/8.07 × 10^{-2} **«**mol**»** ***OR*** mass of Na_2O_2 theoretical yield **«**= $\frac{5.00\text{ g}}{61.98\text{ g mol}^{-1}}$ × 77.98 g mol^{-1}**»** = 6.291 **«**g**»** ✓ % yield **«**= $\frac{5.50\text{ g}}{6.291\text{ g}}$ × 100**»** ***OR*** **«**$\frac{0.0705}{0.0807}$ x 100**»** = 87.4 **«**%**»** ✓	*Award **[2]** for correct final answer.*	**2**

3.	**g**	**i**	$\Sigma\Delta H_f$ products = 2 × (–1130.7) / –2261.4 **«**kJ**»** ✓ $\Sigma\Delta H_f$ reactants = 2 × (–510.9) + 2 × (–393.5) / –1808.8 **«**kJ**»** ✓ ΔH = **«**$\Sigma\Delta H_f$ products – $\Sigma\Delta H_f$ reactants = –2261.4 –(–1808.8) =**»** –452.6 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for "+ 452.6 «kJ»".*	**3**
3.	**g**	**ii**	only valid for covalent bonds ***OR*** only valid in gaseous state ✓		**1**
3.	**h**		bond in O_3 has lower enthalpy ***AND*** bond order is 1.5 **«**not 2**»** ✓	*Accept "bond in ozone is longer".*	**1**
3.	**k**		IV ✓		**1**